

Corso di Scienza delle Costruzioni

**IL MEZZO CONTINUO
ANALISI DELLA DEFORMAZIONE**

ANALISI DELLA DEFORMAZIONE

.

ANALISI DELLA DEFORMAZIONE

CONTINUO : $\left\{ \begin{array}{l} \text{SOLIDO 3D DEFORMABILE} \\ \text{i cui punti materiali siano in} \\ \text{corrispondenza BIUNIVUCA con lo spazio euclideo} \end{array} \right.$

Un corpo è continuo se ogni punto materiale è un punto di accumulazione

N.B. La materia è continua solo ad una mesoscala.

• corpo $\mathcal{B}$ = insieme di punti materiali descritti rispetto a un sistema di coordinate

PAROLE CHIAVE : • continuità

• configurazione del corpo

ANALISI DELLA DEFORMAZIONE

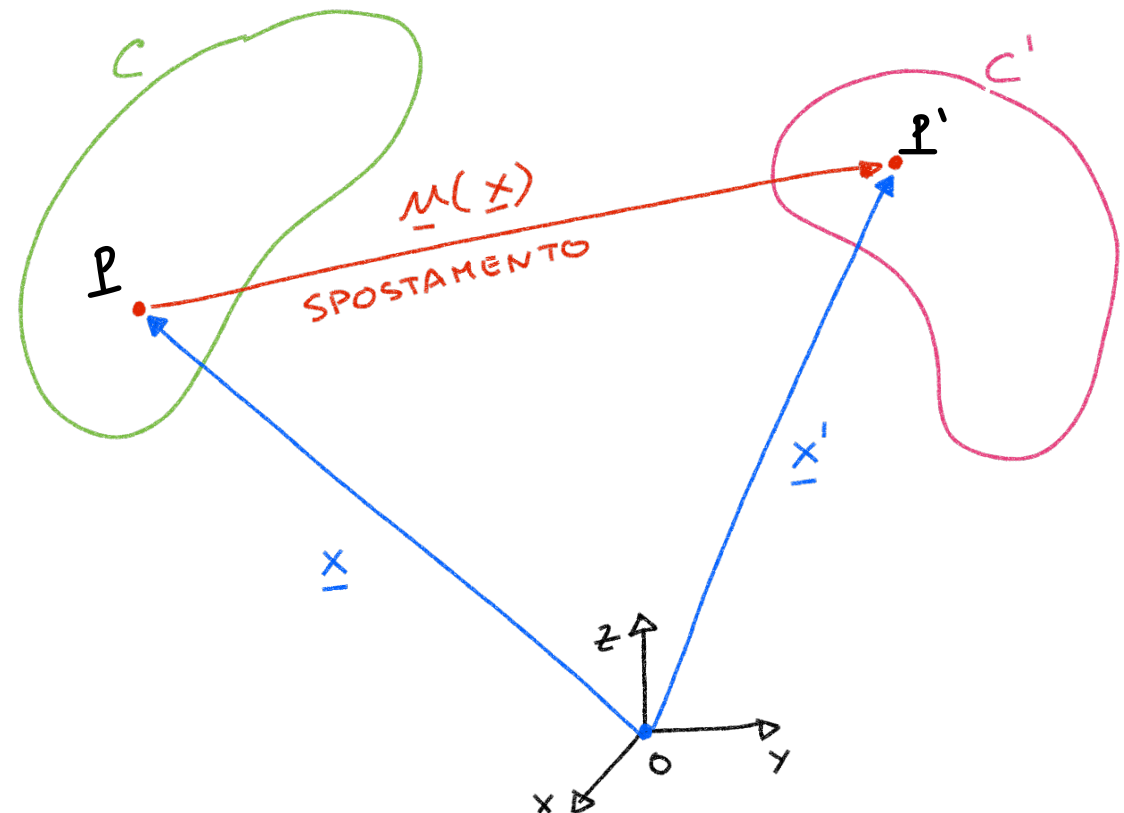
CONTINUO : $\left\{ \begin{array}{l} \text{SOLIDO 3D DEFORMABILE} \\ \text{i cui punti materiali siano in} \\ \text{corrispondenza BIUNIVOCA con lo spazio euclideo} \end{array} \right.$

STUDIO : $\left\{ \begin{array}{l} \text{DEFINIRE UN MODELLO MATEMATICO in grado di} \\ \text{DESCRIVERE LA RISPOSTA STRUTTURALE di un continuo,} \\ \text{IN EQUILIBRIO sotto assegnate azioni esterne.} \end{array} \right.$

ANALISI DELLA DEFORMAZIONE

CONTINUO : $\left\{ \begin{array}{l} \text{SOLIDO 3D DEFORMABILE} \\ \text{i cui punti materiali siano in} \\ \text{corrispondenza BIUNIVUCA con lo spazio euclideo} \end{array} \right.$

STUDIO : $\left\{ \begin{array}{l} \text{DEFINIRE UN MODELLO MATEMATICO in grado di} \\ \text{DESCRIVERE LA RISPOSTA STRUTTURALE di un continuo,} \\ \text{IN EQUILIBRIO sotto assegnate azioni esterne.} \end{array} \right.$



ANALISI DELLA DEFORMAZIONE

CONTINUO: $\left\{ \begin{array}{l} \text{SOLIDO 3D DEFORMABILE} \\ \text{i cui punti materiali siano in} \\ \text{corrispondenza BIUNIVUCA con lo spazio euclideo} \end{array} \right.$

STUDIO: $\left\{ \begin{array}{l} \text{DEFINIRE UN MODELLO MATEMATICO in grado di} \\ \text{DESCRIVERE LA RISPOSTA STRUTTURALE di un continuo,} \\ \text{IN EQUILIBRIO sotto assegnate azioni esterne.} \end{array} \right.$

PROCESSO DEFORMATIVO: $\left\{ \begin{array}{l} \text{CAMBIAMENTO DI} \\ \text{CONFIGURAZIONE} \\ \text{DEL CORPO } \mathcal{B} \end{array} \right.$

$$\left. \begin{array}{l} \underline{x} = [x, y, z]^T \\ \underline{x}' = [x', y', z']^T \end{array} \right\} \underline{u}(\underline{x}) = \underline{x}' - \underline{x} = [u, v, w]^T$$

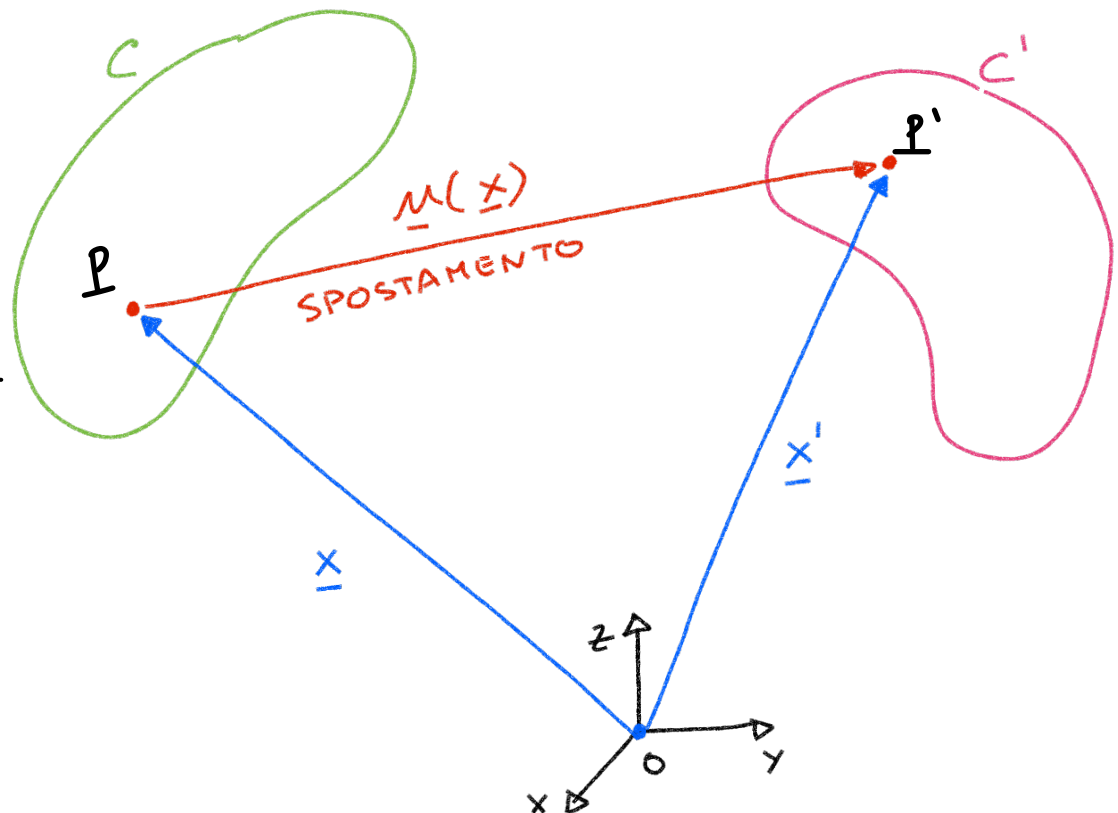
DIFFERENZA CONTINUA

↳ $\underline{x}' = \underline{x} + \underline{u}(\underline{x})$

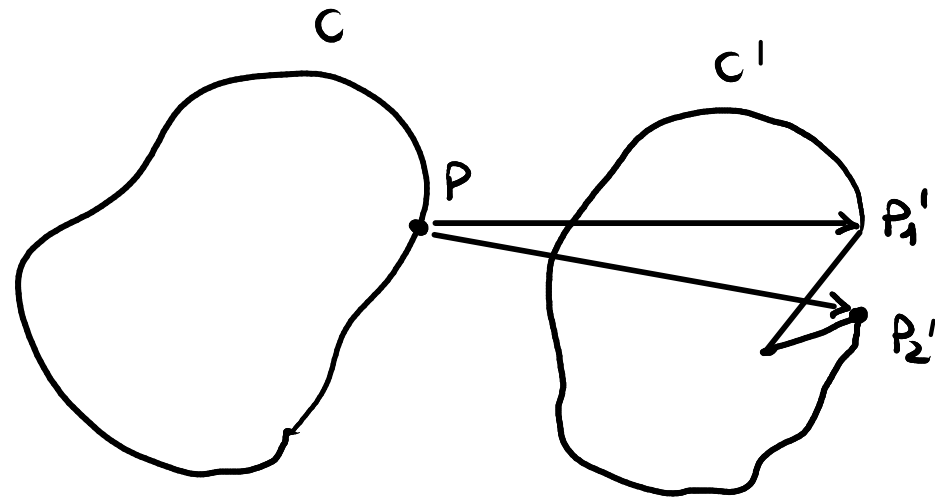
Hp. SPOSTAMENTI
INFINITESIMI

Per passare da C a C' assegno un campo di spostamento $\underline{u}$
MA

Esso non può essere qualsiasi!



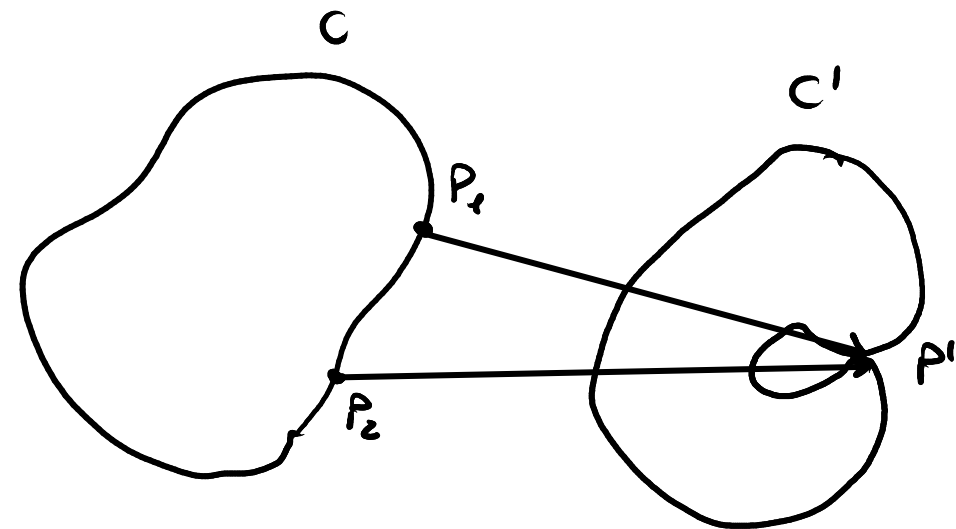
ANALISI DELLA DEFORMAZIONE



Non u devono essere lacerazioni



u deve essere iniettiva su C



Non u devono essere compenetrazioni



u⁻¹ deve essere iniettiva su C'

Se P e Q sono due punti in C vicini tra loro, i loro punti trasformati P' e Q' dovranno continuare ad essere vicini in C'

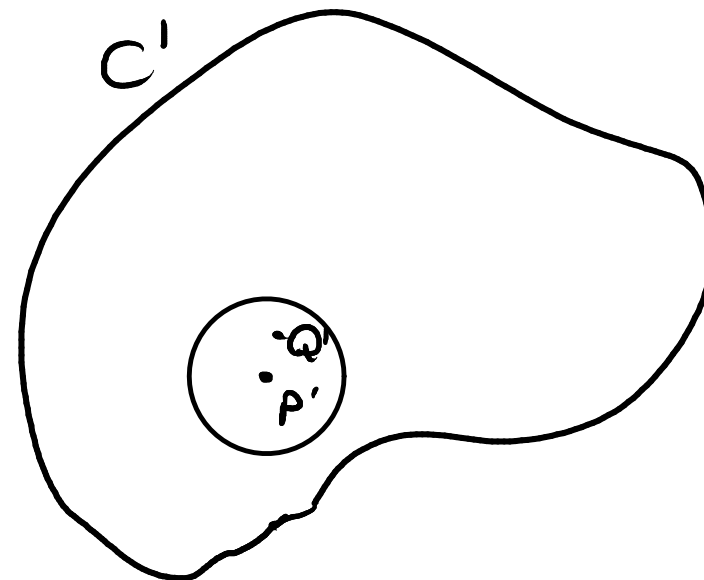
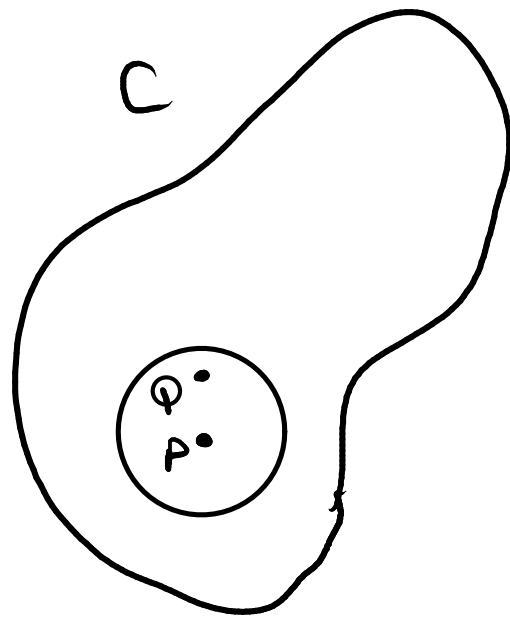
$$\lim_{Q \rightarrow P} \underline{u}(Q) = \underline{u}(P) \quad \Rightarrow \quad \underline{u}(\underline{x}) \text{ deve essere continua su } C$$

Tutti questi requisiti sono soddisfatti se

$$\underline{u}(\underline{x}) \in C^1(C)$$

ANALISI DELLA DEFORMAZIONE

Cinematica degli interni



ANALISI DELLA DEFORMAZIONE

TENSORE DELLA DEFORMAZIONE :

ANALISI DELLA DEFORMAZIONE

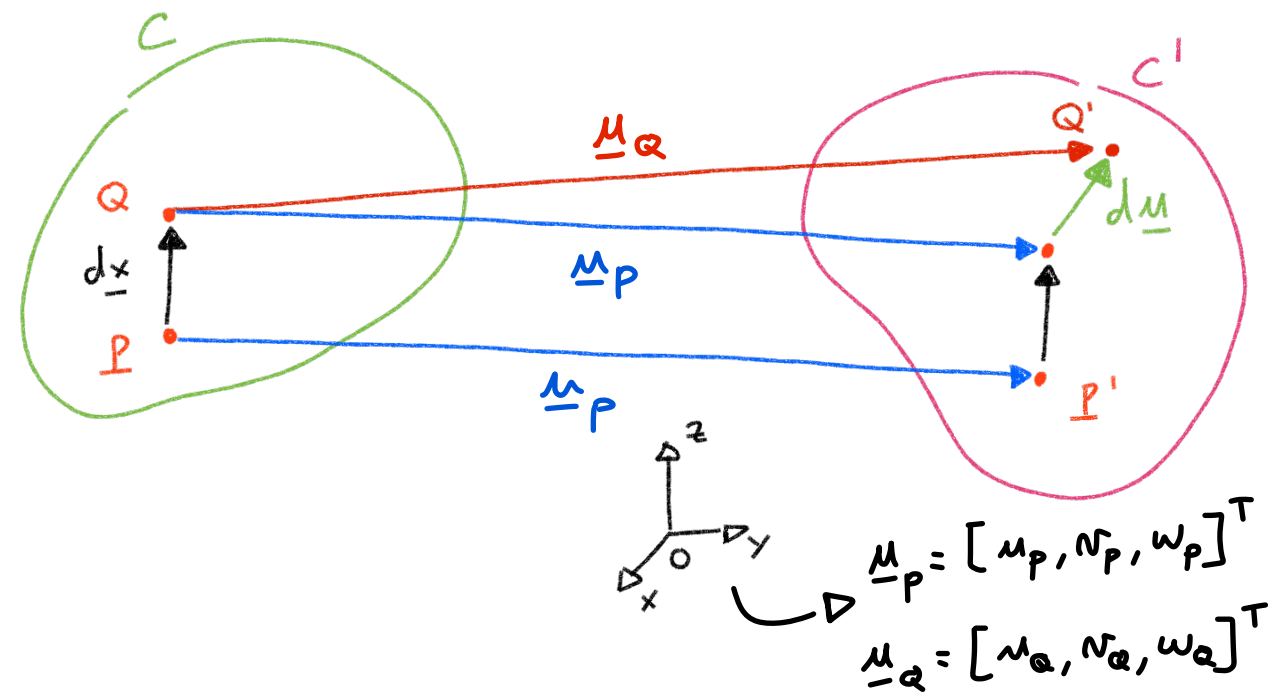
TENSORE DELLA DEFORMAZIONE :

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

ANALISI DELLA DEFORMAZIONE

TENSORE DELLA DEFORMAZIONE:

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

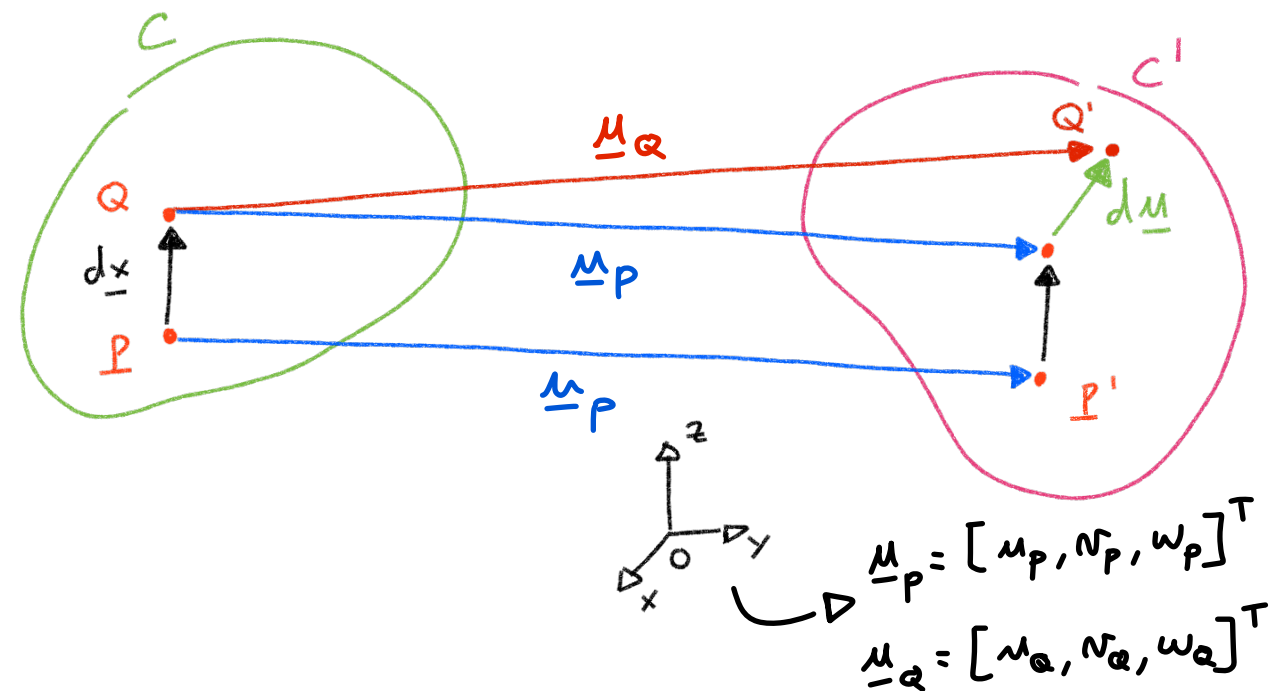


ANALISI DELLA DEFORMAZIONE

TENSORE DELLA DEFORMAZIONE:

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

$$\rightarrow \underline{u}_Q = \underline{u}_P + d\underline{u}(z)$$



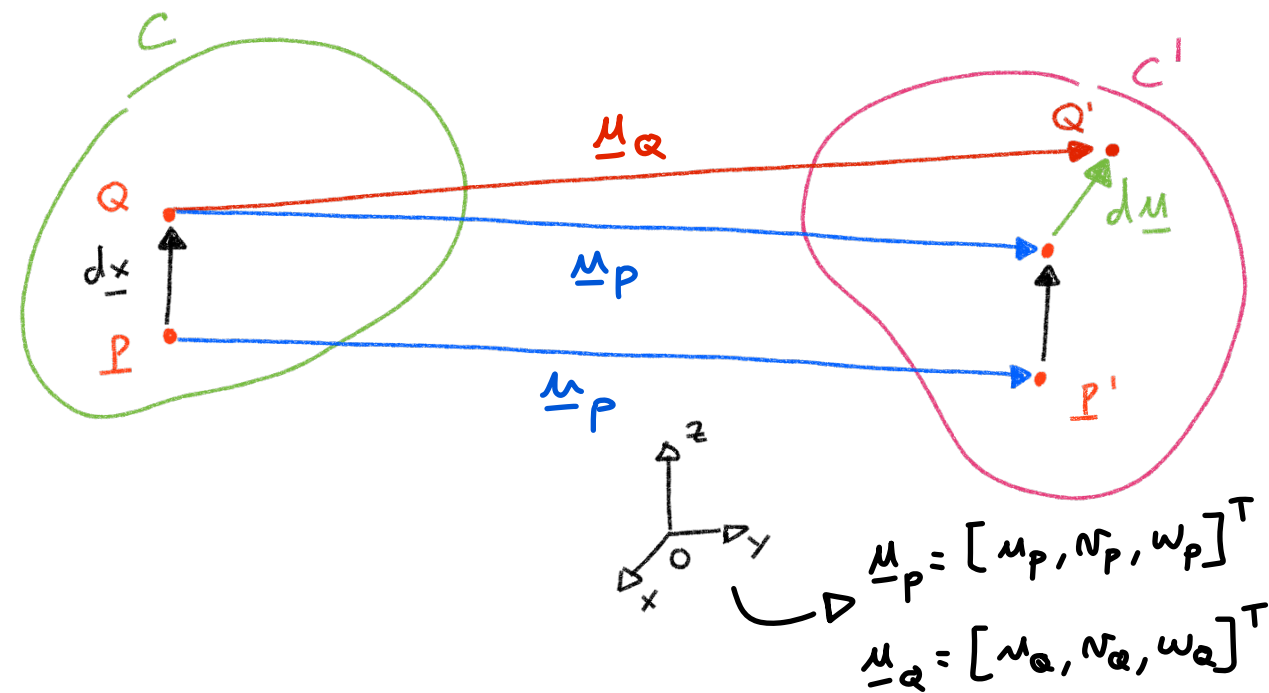
ANALISI DELLA DEFORMAZIONE

TENSORE DELLA DEFORMAZIONE:

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

$$\underline{u}_Q = \underline{u}_P + d\underline{u}(z)$$

$$d\underline{u} = \begin{bmatrix} du \\ dv \\ dw \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \\ \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz \\ \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \frac{\partial w}{\partial z} dz \end{bmatrix}$$



ANALISI DELLA DEFORMAZIONE

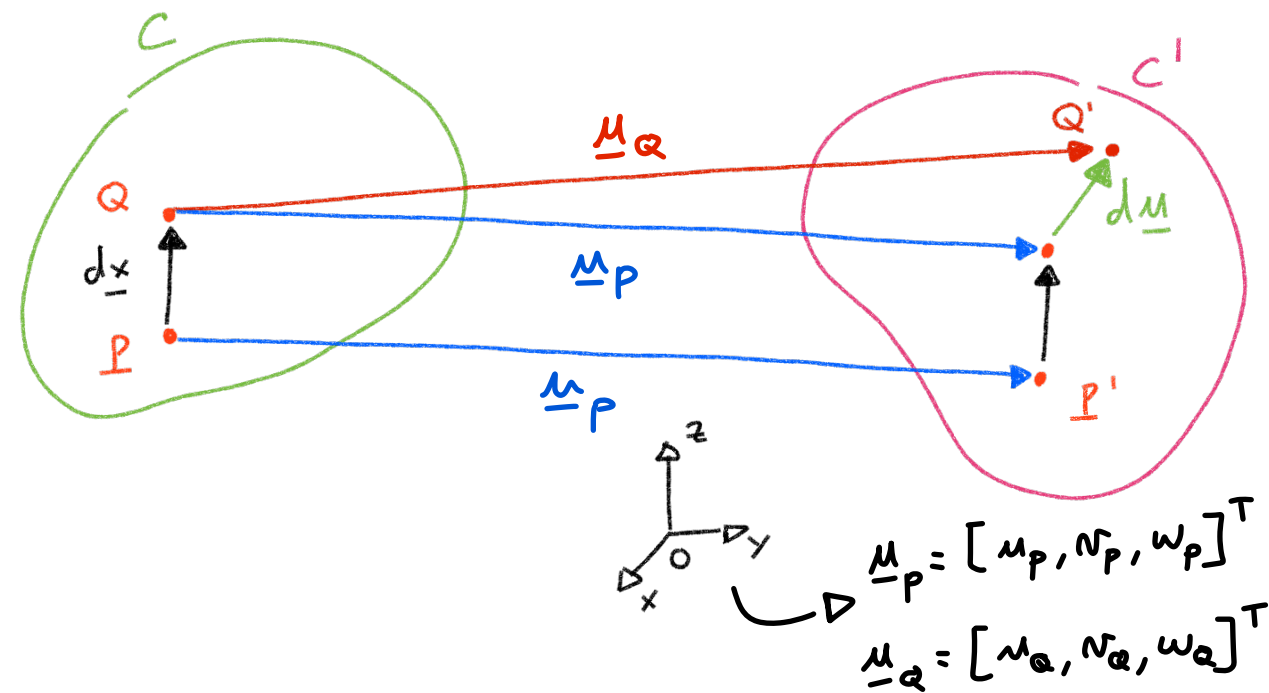
TENSORE DELLA DEFORMAZIONE:

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

$$\rightarrow \underline{\underline{u}}_Q = \underline{\underline{u}}_P + d\underline{\underline{u}}(x)$$

$$d\underline{\underline{u}} = \begin{bmatrix} du \\ dv \\ dw \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \\ \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz \\ \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \frac{\partial w}{\partial z} dz \end{bmatrix}$$

$$\begin{bmatrix} u_Q \\ v_Q \\ w_Q \end{bmatrix} = \begin{bmatrix} u_P \\ v_P \\ w_P \end{bmatrix} + \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix} \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$



ANALISI DELLA DEFORMAZIONE

TENSORE DELLA DEFORMAZIONE:

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

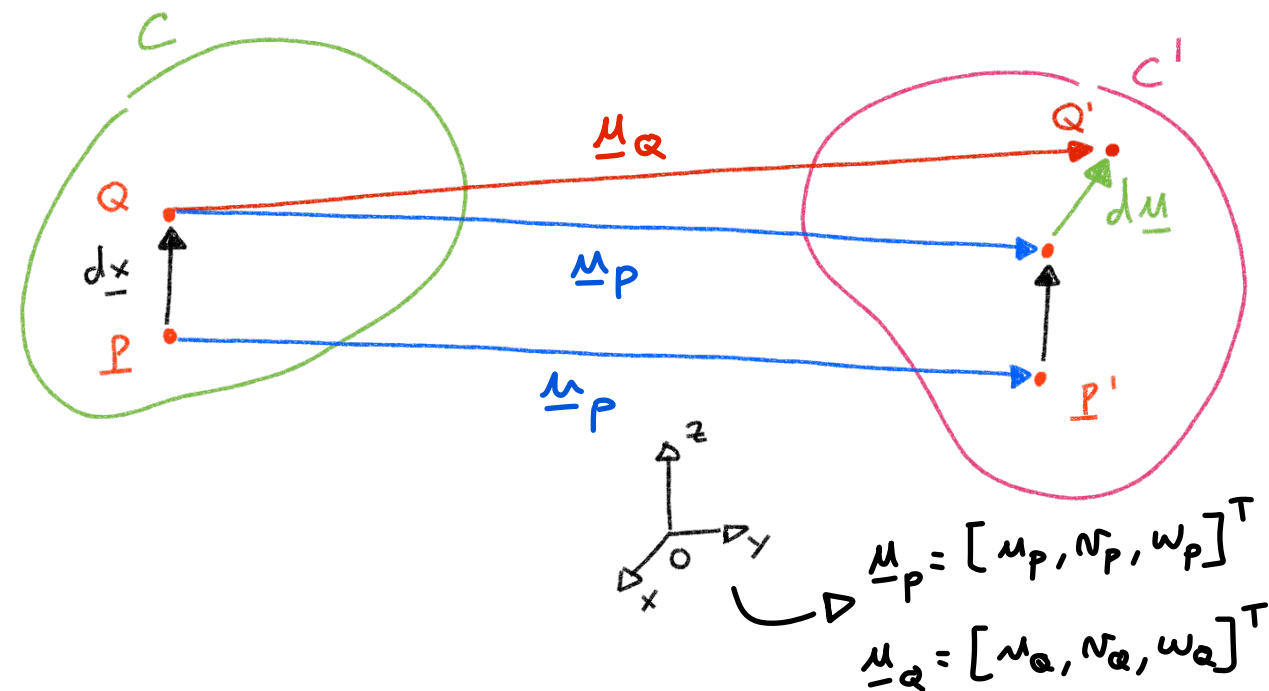
$$\underline{u}_Q = \underline{u}_P + d\underline{u}(\underline{x})$$

$$d\underline{u} = \begin{bmatrix} du \\ dv \\ dw \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \\ \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz \\ \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \frac{\partial w}{\partial z} dz \end{bmatrix}$$

$$\begin{bmatrix} u_Q \\ v_Q \\ w_Q \end{bmatrix} = \begin{bmatrix} u_P \\ v_P \\ w_P \end{bmatrix} + \underbrace{\begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix}}_{\text{MATRICE GRADIENTE DI SPOSTAMENTO}} \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$

MATRICE GRADIENTE
DI SPOSTAMENTO

$$F = \frac{\partial \underline{u}(\underline{x})}{\partial \underline{x}}$$



ANALISI DELLA DEFORMAZIONE

TENSORE DELLA DEFORMAZIONE:

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

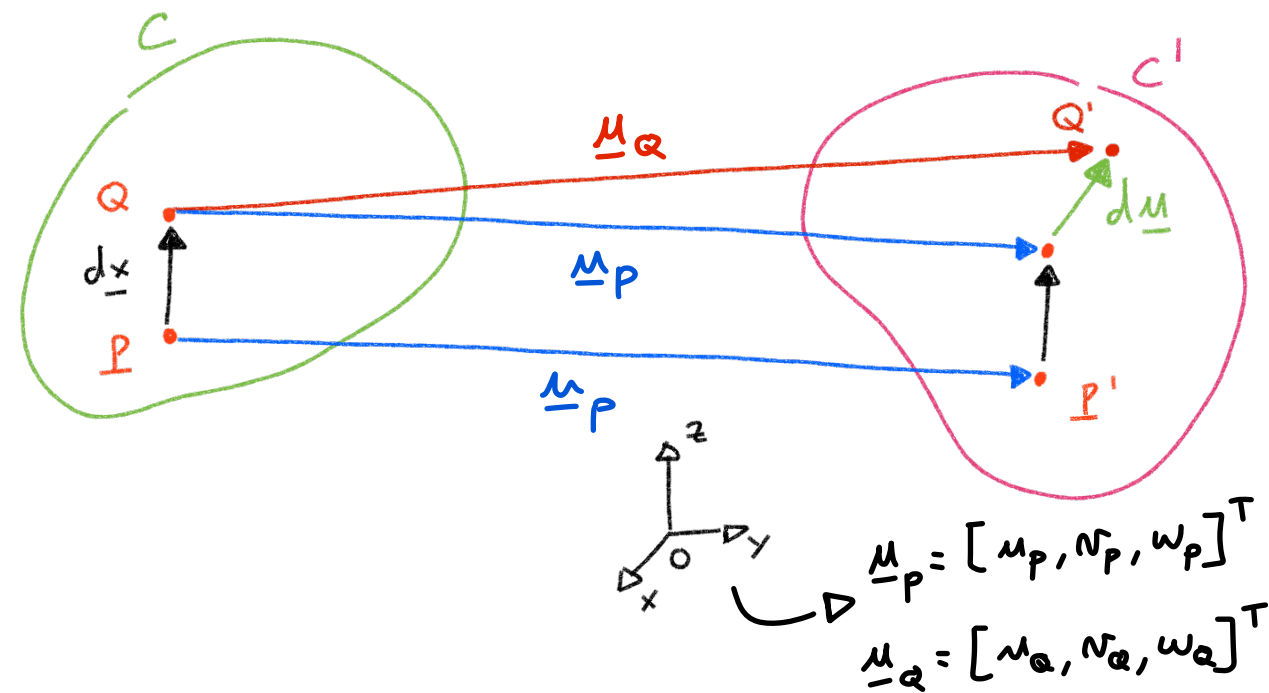
$$\underline{u}_Q = \underline{u}_P + d\underline{u}(x)$$

$$d\underline{u} = \begin{bmatrix} du \\ dv \\ dw \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \\ \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz \\ \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \frac{\partial w}{\partial z} dz \end{bmatrix}$$

$$\begin{bmatrix} u_Q \\ v_Q \\ w_Q \end{bmatrix} = \begin{bmatrix} u_P \\ v_P \\ w_P \end{bmatrix} + \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix} \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$

MATRICE GRADIENTE
DI SPOSTAMENTO

$$\underline{F} = \frac{\partial \underline{u}(x)}{\partial \underline{x}}$$



$$\underline{u}_Q = \underline{u}_P + \underline{F} d\underline{x}$$

ANALISI DELLA DEFORMAZIONE

TENSORE DELLA DEFORMAZIONE:

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

$$\underline{u}_Q = \underline{u}_P + d\underline{u}(x)$$

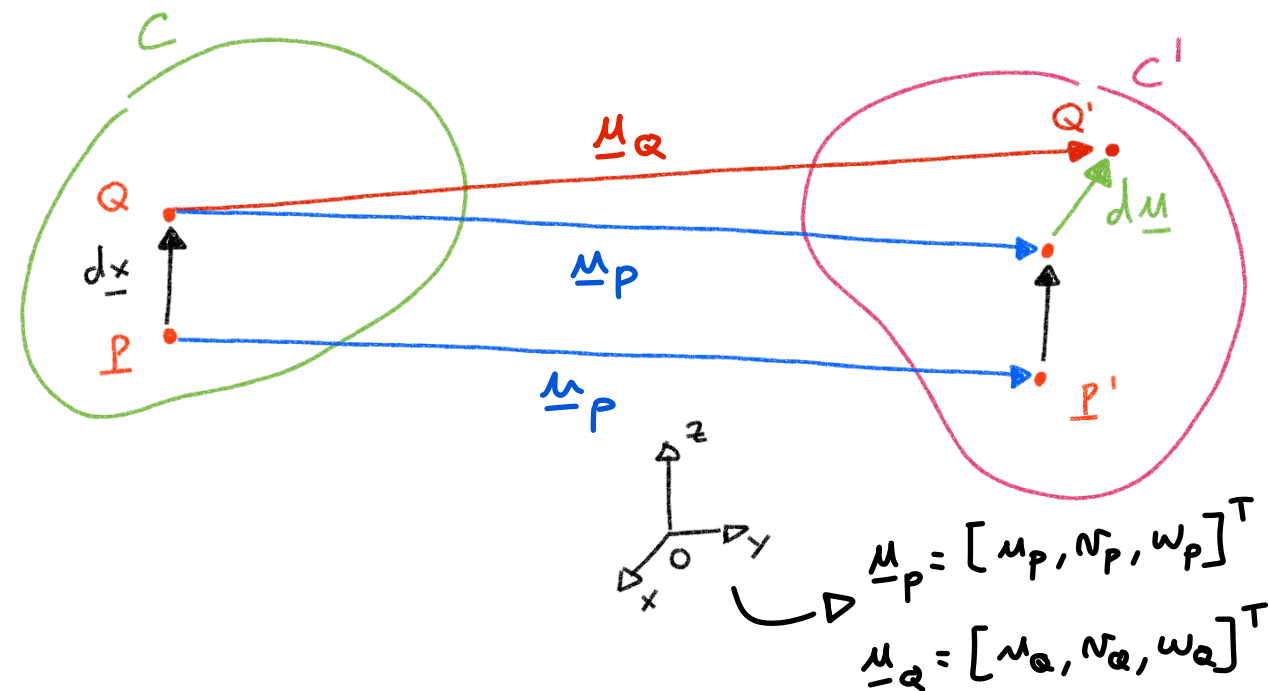
$$d\underline{u} = \begin{bmatrix} du \\ dv \\ dw \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \\ \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz \\ \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \frac{\partial w}{\partial z} dz \end{bmatrix}$$

$$\begin{bmatrix} u_Q \\ v_Q \\ w_Q \end{bmatrix} = \begin{bmatrix} u_P \\ v_P \\ w_P \end{bmatrix} + \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix} \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix} \rightarrow$$

MATRICE GRADIENTE
DI SPOSTAMENTO

$$\underline{F} = \underline{E} + \underline{\Omega}$$

$$\underline{F} = \frac{\partial \underline{u}(x)}{\partial \underline{x}}$$



$$\underline{u}_Q = \underline{u}_P + \underline{F} d\underline{x}$$

ANALISI DELLA DEFORMAZIONE

TENSORE DELLA DEFORMAZIONE:

$$\left. \begin{array}{l} P(x, y, z) \\ Q(x+dx, y+dy, z+dz) \end{array} \right\} \overline{PQ} = d\underline{x}$$

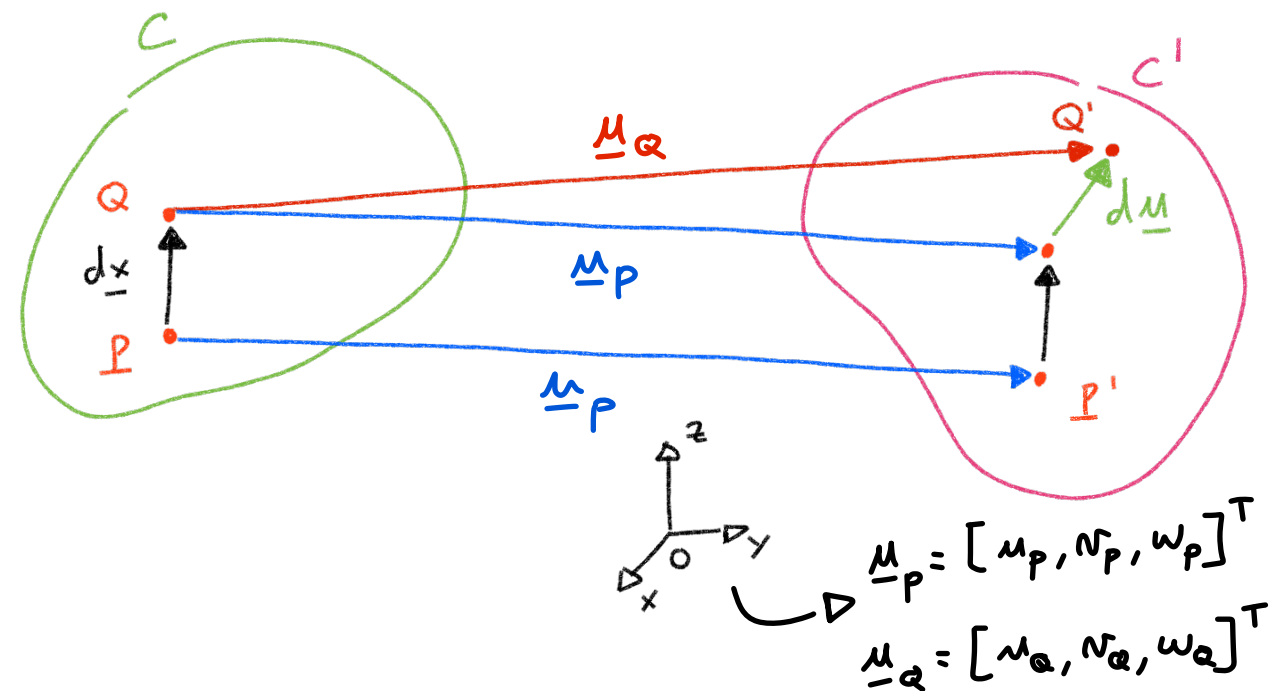
$$\underline{u}_Q = \underline{u}_P + d\underline{u}(x)$$

$$d\underline{u} = \begin{bmatrix} du \\ dv \\ dw \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz \\ \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy + \frac{\partial v}{\partial z} dz \\ \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \frac{\partial w}{\partial z} dz \end{bmatrix}$$

$$\begin{bmatrix} u_Q \\ v_Q \\ w_Q \end{bmatrix} = \begin{bmatrix} u_P \\ v_P \\ w_P \end{bmatrix} + \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix} \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$

MATRICE GRADIENTE
DI SPOSTAMENTO

$$\underline{F} = \underline{E} + \underline{\Omega} \quad \leftarrow \quad \underline{F} = \frac{\partial \underline{u}(x)}{\partial \underline{x}}$$



$$\underline{u}_Q = \underline{u}_P + \underline{F} d\underline{x}$$

$\underline{F}$: MATRICE GRADIENTE DI SPOSTAMENTO
 $\underline{E}$: " DEFORMAZIONE PURA
 (TENSORE DELLA DEFORMAZIONE)
 $\underline{\Omega}$: " ROTAZIONE RIGIDA

ANALISI DELLA DEFORMAZIONE

$\left\{ \begin{array}{l} \underline{F} : \text{MATRICE GRADIENTE DI SPOSTAMENTO} \\ \underline{E} : \text{// DEFORMAZIONE PURA} \\ \quad \text{(TENSORE DELLA DEFORMAZIONE)} \\ \underline{\Omega} : \text{// ROTAZIONE RIGIDA} \end{array} \right.$

ANALISI DELLA DEFORMAZIONE

$\underline{\underline{F}}$: MATRICE GRADIENTE DI SPOSTAMENTO
 $\underline{\underline{E}}$: " DEFORMAZIONE PURA
(TENSORE DELLA DEFORMAZIONE)
 $\underline{\underline{\Omega}}$: " ROTAZIONE RIGIDA

DECOMPOSIZIONE
MATRICIALE

$$\underline{\underline{F}} = \underline{\underline{E}} + \underline{\underline{\Omega}}$$
$$\underline{\underline{E}} = \frac{1}{2} \left[\underline{\underline{F}} + \underline{\underline{F}}^T \right] \quad \text{SYM. } \underline{\underline{F}}$$
$$\underline{\underline{\Omega}} = \frac{1}{2} \left[\underline{\underline{F}} - \underline{\underline{F}}^T \right] \quad \text{ESYM. } \underline{\underline{F}}$$

ANALISI DELLA DEFORMAZIONE

$\underline{\underline{F}}$: MATRICE GRADIENTE DI SPOSTAMENTO
 $\underline{\underline{E}}$: " DEFORMAZIONE PURA
 (TENSORE DELLA DEFORMAZIONE)
 $\underline{\underline{\Omega}}$: " ROTAZIONE RIGIDA

DECOMPOSIZIONE
MATRICIALE

$$\underline{\underline{F}} = \underline{\underline{E}} + \underline{\underline{\Omega}}$$

$$\underline{\underline{E}} = \frac{1}{2} \left[\underline{\underline{F}} + \underline{\underline{F}}^T \right] \quad \text{SYM. } \underline{\underline{F}}$$

$$\underline{\underline{\Omega}} = \frac{1}{2} \left[\underline{\underline{F}} - \underline{\underline{F}}^T \right] \quad \text{ESYM. } \underline{\underline{F}}$$

$$[F] = \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix}; [F^T] = \begin{bmatrix} \partial u / \partial x & \partial v / \partial x & \partial w / \partial x \\ \partial u / \partial y & \partial v / \partial y & \partial w / \partial y \\ \partial u / \partial z & \partial v / \partial z & \partial w / \partial z \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$\underline{\underline{F}}$: MATRICE GRADIENTE DI SPOSTAMENTO
 $\underline{\underline{E}}$: " DEFORMAZIONE PURA
 (TENSORE DELLA DEFORMAZIONE)
 $\underline{\underline{\Omega}}$: " ROTAZIONE RIGIDA

DECOMPOSIZIONE
MATRICIALE

$$\underline{\underline{F}} = \underline{\underline{E}} + \underline{\underline{\Omega}}$$

$$\underline{\underline{E}} = \frac{1}{2} \left[\underline{\underline{F}} + \underline{\underline{F}}^T \right]$$

SYM. $\underline{\underline{F}}$

$$\underline{\underline{\Omega}} = \frac{1}{2} \left[\underline{\underline{F}} - \underline{\underline{F}}^T \right]$$

ESYM. $\underline{\underline{F}}$

$$[F] = \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix}; [F^T] = \begin{bmatrix} \partial u / \partial x & \partial v / \partial x & \partial w / \partial x \\ \partial u / \partial y & \partial v / \partial y & \partial w / \partial y \\ \partial u / \partial z & \partial v / \partial z & \partial w / \partial z \end{bmatrix}$$

$$[E] = \begin{bmatrix} \partial u / \partial x & 1/2 (\partial u / \partial y + \partial v / \partial x) & 1/2 (\partial u / \partial z + \partial w / \partial x) \\ & \partial v / \partial y & 1/2 (\partial v / \partial z + \partial w / \partial y) \\ \text{SYM} & & \partial w / \partial z \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$\underline{\underline{F}}$: MATRICE GRADIENTE DI SPOSTAMENTO
 $\underline{\underline{E}}$: " DEFORMAZIONE PURA
 (TENSORE DELLA DEFORMAZIONE)
 $\underline{\underline{\Omega}}$: " ROTAZIONE RIGIDA

DECOMPOSIZIONE
MATRICIALE

$$\underline{\underline{F}} = \underline{\underline{E}} + \underline{\underline{\Omega}}$$

$$\underline{\underline{E}} = \frac{1}{2} \left[\underline{\underline{F}} + \underline{\underline{F}}^T \right]$$

SYM. $\underline{\underline{F}}$

$$\underline{\underline{\Omega}} = \frac{1}{2} \left[\underline{\underline{F}} - \underline{\underline{F}}^T \right]$$

ESYM. $\underline{\underline{F}}$

$$[F] = \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix}; [F^T] = \begin{bmatrix} \partial u / \partial x & \partial v / \partial x & \partial w / \partial x \\ \partial u / \partial y & \partial v / \partial y & \partial w / \partial y \\ \partial u / \partial z & \partial v / \partial z & \partial w / \partial z \end{bmatrix}$$

$$[E] = \begin{bmatrix} \partial u / \partial x & 1/2 (\partial u / \partial y + \partial v / \partial x) & 1/2 (\partial u / \partial z + \partial w / \partial x) \\ & \partial v / \partial y & 1/2 (\partial v / \partial z + \partial w / \partial y) \\ \text{SYM} & & \partial w / \partial z \end{bmatrix}$$

$$[\Omega] = \begin{bmatrix} 0 & 1/2 (\partial u / \partial y - \partial v / \partial x) & 1/2 (\partial u / \partial z - \partial w / \partial x) \\ 1/2 (\partial v / \partial x - \partial u / \partial y) & 0 & 1/2 (\partial v / \partial z - \partial w / \partial y) \\ 1/2 (\partial w / \partial x - \partial u / \partial z) & 1/2 (\partial w / \partial y - \partial v / \partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$\underline{\underline{F}}$: MATRICE GRADIENTE DI SPOSTAMENTO
 $\underline{\underline{E}}$: " DEFORMAZIONE PURA
 (TENSORE DELLA DEFORMAZIONE)
 $\underline{\underline{\Omega}}$: " ROTAZIONE RIGIDA

DECOMPOSIZIONE
MATRICIALE

$$\underline{\underline{F}} = \underline{\underline{E}} + \underline{\underline{\Omega}}$$

$$\underline{\underline{E}} = \frac{1}{2} \left[\underline{\underline{F}} + \underline{\underline{F}}^T \right]$$

SYM. $\underline{\underline{F}}$

$$\underline{\underline{\Omega}} = \frac{1}{2} \left[\underline{\underline{F}} - \underline{\underline{F}}^T \right]$$

ESYM. $\underline{\underline{F}}$

$$[F] = \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix}; [F^T] = \begin{bmatrix} \partial u / \partial x & \partial v / \partial x & \partial w / \partial x \\ \partial u / \partial y & \partial v / \partial y & \partial w / \partial y \\ \partial u / \partial z & \partial v / \partial z & \partial w / \partial z \end{bmatrix}$$

→

$$\underline{\underline{u}}_Q = \underline{\underline{u}}_P + \underline{\underline{F}} d\underline{\underline{x}}$$

$$[E] = \begin{bmatrix} \partial u / \partial x & 1/2 (\partial u / \partial y + \partial v / \partial x) & 1/2 (\partial u / \partial z + \partial w / \partial x) \\ & \partial v / \partial y & 1/2 (\partial v / \partial z + \partial w / \partial y) \\ \text{SYM} & & \partial w / \partial z \end{bmatrix}$$

$$[\Omega] = \begin{bmatrix} 0 & 1/2 (\partial u / \partial y - \partial v / \partial x) & 1/2 (\partial u / \partial z - \partial w / \partial x) \\ 1/2 (\partial v / \partial x - \partial u / \partial y) & 0 & 1/2 (\partial v / \partial z - \partial w / \partial y) \\ 1/2 (\partial w / \partial x - \partial u / \partial z) & 1/2 (\partial w / \partial y - \partial v / \partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$\underline{\underline{F}}$: MATRICE GRADIENTE DI SPOSTAMENTO
 $\underline{\underline{E}}$: " DEFORMAZIONE PURA
 (TENSORE DELLA DEFORMAZIONE)
 $\underline{\underline{\Omega}}$: " ROTAZIONE RIGIDA

DECOMPOSIZIONE
MATRICIALE

$$\underline{\underline{F}} = \underline{\underline{E}} + \underline{\underline{\Omega}}$$

$$\underline{\underline{E}} = \frac{1}{2} [\underline{\underline{F}} + \underline{\underline{F}}^T]$$

SYM. $\underline{\underline{F}}$

$$\underline{\underline{\Omega}} = \frac{1}{2} [\underline{\underline{F}} - \underline{\underline{F}}^T]$$

ESYM. $\underline{\underline{F}}$

$$[F] = \begin{bmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{bmatrix}; [F^T] = \begin{bmatrix} \partial u / \partial x & \partial v / \partial x & \partial w / \partial x \\ \partial u / \partial y & \partial v / \partial y & \partial w / \partial y \\ \partial u / \partial z & \partial v / \partial z & \partial w / \partial z \end{bmatrix}$$

$$[E] = \begin{bmatrix} \partial u / \partial x & 1/2 (\partial u / \partial y + \partial v / \partial x) & 1/2 (\partial u / \partial z + \partial w / \partial x) \\ & \partial v / \partial y & 1/2 (\partial v / \partial z + \partial w / \partial y) \\ & & \partial w / \partial z \end{bmatrix}$$

SYM

$$[\Omega] = \begin{bmatrix} 0 & 1/2 (\partial u / \partial y - \partial v / \partial x) & 1/2 (\partial u / \partial z - \partial w / \partial x) \\ 1/2 (\partial v / \partial x - \partial u / \partial y) & 0 & 1/2 (\partial v / \partial z - \partial w / \partial y) \\ 1/2 (\partial w / \partial x - \partial u / \partial z) & 1/2 (\partial w / \partial y - \partial v / \partial z) & 0 \end{bmatrix}$$

$$\underline{\underline{u}}_Q = \underline{\underline{u}}_P + \underline{\underline{F}} d_x$$

$$\underline{\underline{u}}_Q = \underline{\underline{u}}_P + \underline{\underline{E}} d_x + \underline{\underline{\Omega}} d_x$$

SPOSTAMENTO DI UN PUNTO Q
NELL'INTORNO DI P

ANALISI DELLA DEFORMAZIONE

:

ANALISI DELLA DEFORMAZIONE

$$\underline{\mu}_Q = \underline{\mu}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

:

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

ROTAZIONE
RIGIDA

:

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

ROTAZIONE
RIGIDA

:

$$[\underline{\Omega}] = \begin{bmatrix} 0 & 1/2(\partial u/\partial y - \partial v/\partial x) & 1/2(\partial u/\partial z - \partial w/\partial x) \\ 1/2(\partial v/\partial x - \partial u/\partial y) & 0 & 1/2(\partial v/\partial z - \partial w/\partial y) \\ 1/2(\partial w/\partial x - \partial u/\partial z) & 1/2(\partial w/\partial y - \partial v/\partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

ROTAZIONE
RIGIDA

:

$$[\underline{\Omega}] = \begin{bmatrix} 0 & 1/2(\partial u/\partial y - \partial v/\partial x) & 1/2(\partial u/\partial z - \partial w/\partial x) \\ 1/2(\partial v/\partial x - \partial u/\partial y) & 0 & 1/2(\partial v/\partial z - \partial w/\partial y) \\ 1/2(\partial w/\partial x - \partial u/\partial z) & 1/2(\partial w/\partial y - \partial v/\partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\bar{E}} d\underline{x} + \underline{\bar{\Omega}} d\underline{x}$$

ROTAZIONE
RIGIDA

POSTI:

$$\begin{cases} \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & \frac{1}{2}(\partial u / \partial y - \partial v / \partial x) & \frac{1}{2}(\partial u / \partial z - \partial w / \partial x) \\ \frac{1}{2}(\partial v / \partial x - \partial u / \partial y) & 0 & \frac{1}{2}(\partial v / \partial z - \partial w / \partial y) \\ \frac{1}{2}(\partial w / \partial x - \partial v / \partial z) & \frac{1}{2}(\partial w / \partial y - \partial u / \partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

ROTAZIONE
RIGIDA

POSTI:

$$\begin{cases} \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & \frac{1}{2}(\partial u / \partial y - \partial v / \partial x) & \frac{1}{2}(\partial u / \partial z - \partial w / \partial x) \\ \frac{1}{2}(\partial w / \partial x - \partial u / \partial y) & 0 & \frac{1}{2}(\partial v / \partial z - \partial w / \partial y) \\ \frac{1}{2}(\partial w / \partial x - \partial u / \partial y) & \frac{1}{2}(\partial w / \partial y - \partial v / \partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

ROTAZIONE
RIGIDA

$$\underline{\omega} = [\omega_x \ \omega_y \ \omega_z]^T$$

POSTI:

$$\begin{cases} \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & \frac{1}{2}(\partial u / \partial y - \partial v / \partial x) & \frac{1}{2}(\partial u / \partial z - \partial w / \partial x) \\ \frac{1}{2}(\partial v / \partial x - \partial u / \partial y) & 0 & \frac{1}{2}(\partial v / \partial z - \partial w / \partial y) \\ \frac{1}{2}(\partial w / \partial x - \partial v / \partial z) & \frac{1}{2}(\partial w / \partial y - \partial v / \partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

ROTAZIONE
RIGIDA

$$\underline{w} = [w_x \ w_y \ w_z]^T$$

POSTI:

$$\begin{cases} w_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ w_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ w_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$[\underline{\underline{\Omega}}] = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

$$[\underline{\underline{\Omega}}] = \begin{bmatrix} 0 & 1/2(\partial u/\partial y - \partial v/\partial x) & 1/2(\partial u/\partial z - \partial w/\partial x) \\ 1/2(\partial w/\partial x - \partial u/\partial y) & 0 & 1/2(\partial v/\partial z - \partial w/\partial y) \\ 1/2(\partial w/\partial x - \partial u/\partial y) & 1/2(\partial w/\partial y - \partial v/\partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

MISURE DI
DEFORMAZIONE

ROTAZIONE
RIGIDA

POSTI:

$$\begin{cases} \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$\underline{\omega} = [\omega_x \ \omega_y \ \omega_z]^T$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & 1/2(\partial u/\partial y - \partial v/\partial x) & 1/2(\partial u/\partial z - \partial w/\partial x) \\ 1/2(\partial v/\partial x - \partial u/\partial y) & 0 & 1/2(\partial v/\partial z - \partial w/\partial y) \\ 1/2(\partial w/\partial x - \partial v/\partial z) & 1/2(\partial w/\partial y - \partial v/\partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

MISURE DI
DEFORMAZIONE

ROTAZIONE
RIGIDA

POSTI:

$$\begin{cases} \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$\underline{\omega} = [\omega_x \ \omega_y \ \omega_z]^T$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

$$[E] = \begin{bmatrix} \partial u / \partial x & 1/2 (\partial u / \partial y + \partial v / \partial x) & 1/2 (\partial u / \partial z + \partial w / \partial x) \\ & \partial v / \partial y & 1/2 (\partial v / \partial z + \partial w / \partial y) \\ \text{SYM} & & \partial w / \partial z \end{bmatrix}$$

$$[\Omega] = \begin{bmatrix} 0 & 1/2 (\partial u / \partial y - \partial v / \partial x) & 1/2 (\partial u / \partial z - \partial w / \partial x) \\ 1/2 (\partial v / \partial x - \partial u / \partial y) & 0 & 1/2 (\partial v / \partial z - \partial w / \partial y) \\ 1/2 (\partial w / \partial x - \partial v / \partial z) & 1/2 (\partial w / \partial y - \partial u / \partial z) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

MISURE DI
DEFORMAZIONE

ROTAZIONE
RIGIDA

$$\underline{w} = [w_x \ w_y \ w_z]^T$$

POSTI:

$$\begin{cases} w_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ w_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ w_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

DILATAZIONI : $\varepsilon_x = \frac{\partial u}{\partial x}$; $\varepsilon_y = \frac{\partial v}{\partial y}$; $\varepsilon_z = \frac{\partial w}{\partial z}$

$$[E] = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2}(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}) & \frac{1}{2}(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}) \\ \text{SYM} & \frac{\partial v}{\partial y} & \frac{1}{2}(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}) \\ & & \frac{\partial w}{\partial z} \end{bmatrix}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & \frac{1}{2}(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}) & \frac{1}{2}(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}) \\ \frac{1}{2}(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}) & 0 & \frac{1}{2}(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y}) \\ \frac{1}{2}(\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z}) & \frac{1}{2}(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

MISURE DI
DEFORMAZIONE

ROTAZIONE
RIGIDA

$$\underline{w} = [w_x \ w_y \ w_z]^T$$

POSTI:

$$\begin{cases} w_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ w_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ w_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

DILATAZIONI: $\varepsilon_x = \frac{\partial u}{\partial x}$; $\varepsilon_y = \frac{\partial v}{\partial y}$; $\varepsilon_z = \frac{\partial w}{\partial z}$

SCORRIMENTI ANGOLARI: $\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$; $\gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$; $\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$

$$[E] = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2}(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}) & \frac{1}{2}(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}) \\ \text{SYM} & \frac{\partial v}{\partial y} & \frac{1}{2}(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}) \\ & & \frac{\partial w}{\partial z} \end{bmatrix}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & \frac{1}{2}(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}) & \frac{1}{2}(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}) \\ \frac{1}{2}(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}) & 0 & \frac{1}{2}(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y}) \\ \frac{1}{2}(\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z}) & \frac{1}{2}(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

MISURE DI
DEFORMAZIONE

ROTAZIONE
RIGIDA

POSTI:

$$\begin{cases} w_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ w_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ w_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$\underline{w} = [w_x \ w_y \ w_z]^T$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

DILATAZIONI : $\varepsilon_x = \frac{\partial u}{\partial x}$; $\varepsilon_y = \frac{\partial v}{\partial y}$; $\varepsilon_z = \frac{\partial w}{\partial z}$

SCORRIMENTI ANGOLARI : $\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$; $\gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$; $\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$

$$[\underline{E}] = \begin{bmatrix} \varepsilon_x & \gamma_{xy}/2 & \gamma_{xz}/2 \\ & \varepsilon_y & \gamma_{yz}/2 \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$

$$[\underline{E}] = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2}(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}) & \frac{1}{2}(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}) \\ & \frac{\partial v}{\partial y} & \frac{1}{2}(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}) \\ \text{SYM} & & \frac{\partial w}{\partial z} \end{bmatrix}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & \frac{1}{2}(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}) & \frac{1}{2}(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}) \\ \frac{1}{2}(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}) & 0 & \frac{1}{2}(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y}) \\ \frac{1}{2}(\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z}) & \frac{1}{2}(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}) & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

MISURE DI
DEFORMAZIONE

ROTAZIONE
RIGIDA

POSTI:

$$\begin{cases} \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$\underline{\omega} = [\omega_x \ \omega_y \ \omega_z]^T$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

DILATAZIONI: $\varepsilon_x = \frac{\partial u}{\partial x}$; $\varepsilon_y = \frac{\partial v}{\partial y}$; $\varepsilon_z = \frac{\partial w}{\partial z}$

SCORRIMENTI ANGOLARI: $\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$; $\gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$; $\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$

$$[\underline{E}] = \begin{bmatrix} \varepsilon_x & \gamma_{xy}/2 & \gamma_{xz}/2 \\ & \varepsilon_y & \gamma_{yz}/2 \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

MISURE DI
DEFORMAZIONE

ROTAZIONE
RIGIDA

$$\underline{\underline{\Omega}} = [\omega_x \ \omega_y \ \omega_z]^T$$

$$\text{POSTI: } \begin{cases} \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$[\underline{\underline{\Omega}}] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

DILATAZIONI: $\varepsilon_x = \frac{\partial u}{\partial x}$; $\varepsilon_y = \frac{\partial v}{\partial y}$; $\varepsilon_z = \frac{\partial w}{\partial z}$

SCORRIMENTI ANGOLARI: $\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$; $\gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$; $\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$

$$[\underline{\underline{E}}] = \begin{bmatrix} \varepsilon_x & \gamma_{xy}/2 & \gamma_{xz}/2 \\ & \varepsilon_y & \gamma_{yz}/2 \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$



FORMULA
GENERALE
SPOST. RIGIDO
INFINITESIMO

$$\underline{u}_P = \underline{u}_O + \underline{\underline{\Omega}}_R \underline{x}$$

$$\text{con } [\underline{\underline{\Omega}}_R] = \begin{bmatrix} 0 & -\partial_z & \partial_y \\ \partial_z & 0 & -\partial_x \\ -\partial_y & \partial_x & 0 \end{bmatrix}$$

ANALISI DELLA DEFORMAZIONE

$$\underline{u}_Q = \underline{u}_P + \underline{\underline{E}} d\underline{x} + \underline{\underline{\Omega}} d\underline{x}$$

MISURE DI
DEFORMAZIONE

ROTAZIONE
RIGIDA

$$\underline{w} = [w_x \ w_y \ w_z]^T$$

POSTI:

$$\begin{cases} w_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \\ w_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ w_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$[\underline{\Omega}] = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

DILATAZIONI : $\varepsilon_x = \frac{\partial u}{\partial x}$; $\varepsilon_y = \frac{\partial v}{\partial y}$; $\varepsilon_z = \frac{\partial w}{\partial z}$

SCORRIMENTI ANGOLARI : $\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$; $\gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$; $\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$

$$[\underline{E}] = \begin{bmatrix} \varepsilon_x & \gamma_{xy}/2 & \gamma_{xz}/2 \\ & \varepsilon_y & \gamma_{yz}/2 \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$



FORMULA
GENERALE
SPOST. RIGIDO
INFINITESIMO

$$\underline{u}_P = \underline{u}_O + \underline{\underline{\Omega}}_R \underline{x}$$

con

$$[\underline{\Omega}_R] = \begin{bmatrix} 0 & -\partial_z & \partial_y \\ \partial_z & 0 & -\partial_x \\ -\partial_y & \partial_x & 0 \end{bmatrix}$$

DILATAZIONI

DILATAZIONI

$\overline{\Delta}$: fibra di lunghezza
iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [u, v, 0]^T$ ($w = 0$)

DILATAZIONI

$\overline{PA}$: fibra di lunghezza
iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [u, v, 0]^T$ ($w = 0$)

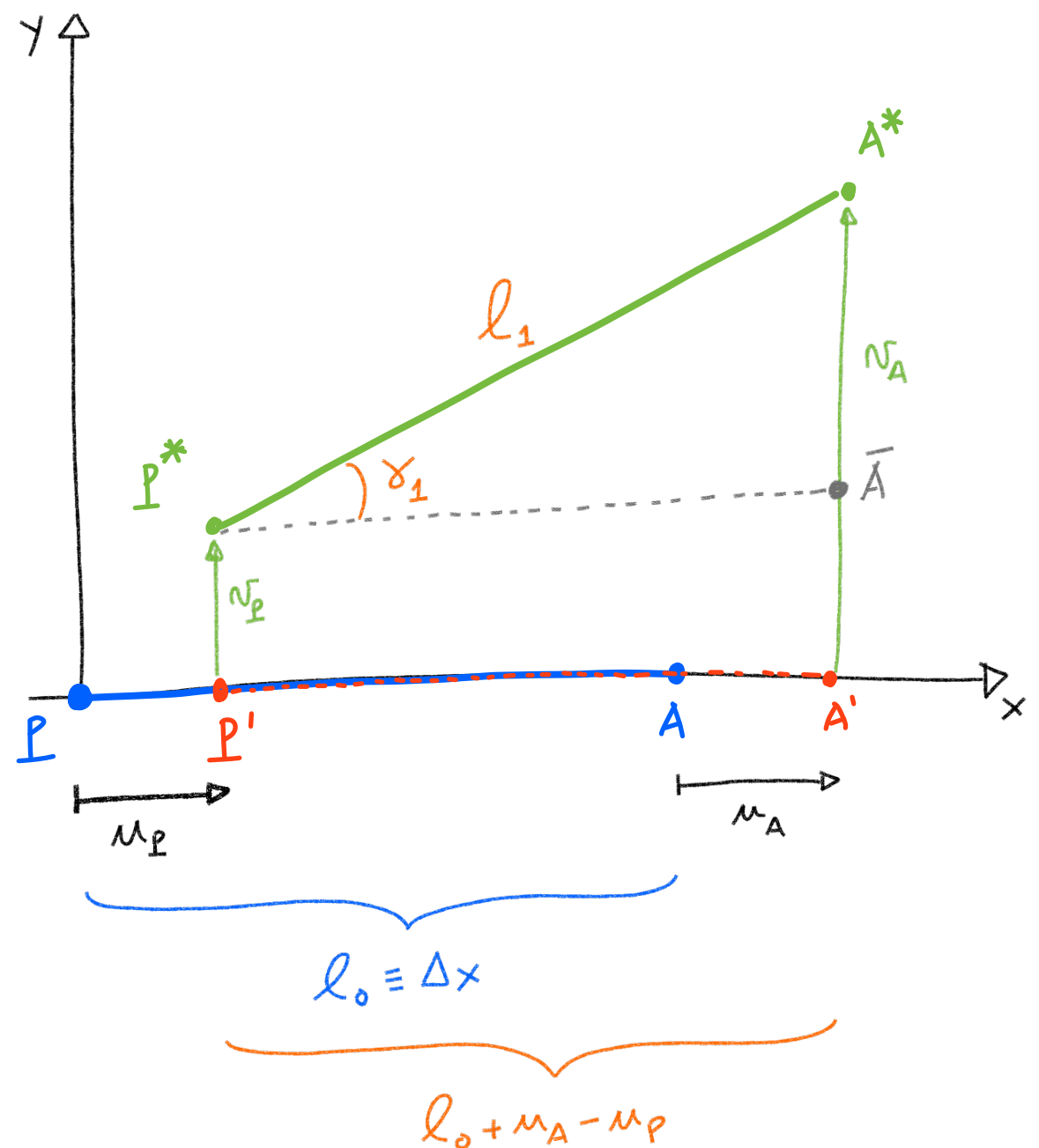
$\overline{P^*A^*}$: fibra deformata di lunghezza l_1

DILATAZIONI

$\overline{PA}$: fibra di lunghezza
iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [\mu, \nu, 0]^T$ ($\omega = 0$)

$\overline{P^*A^*}$: fibra deformata di lunghezza l_1



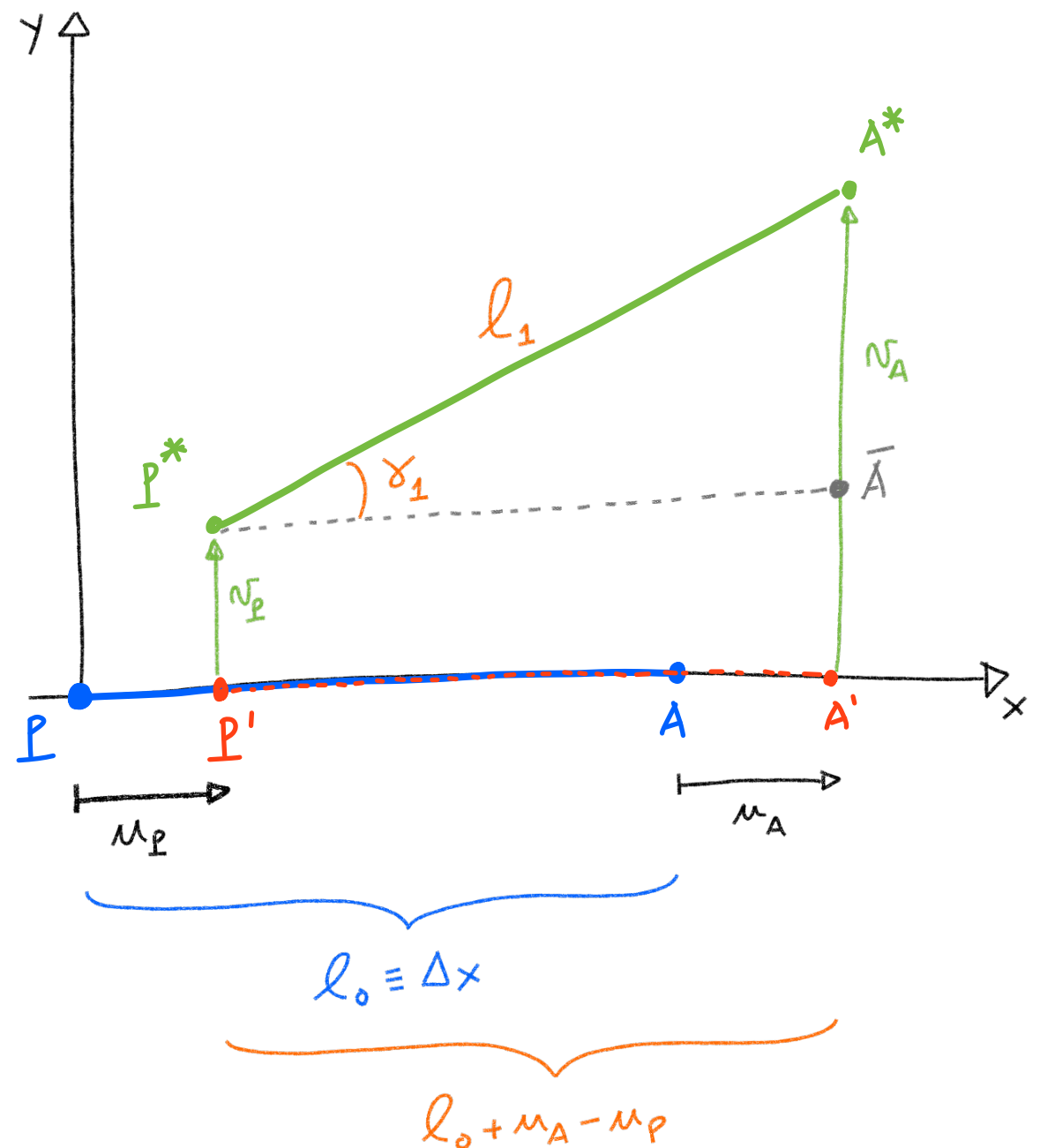
DILATAZIONI

$\overline{PA}$: fibra di lunghezza
iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [\mu, \nu, 0]^T$ ($\omega = 0$)

$\overline{P^*A^*}$: fibra deformata di lunghezza l_1

$$\begin{array}{ll} \underline{P}(x, y, z) & ; A(x + \Delta x, y, z) \\ \mu_P = \mu(x, y, z) & ; \mu_A = \mu(x + \Delta x, y, z) \\ \nu_P = \nu(x, y, z) & ; \nu_A = \nu(x + \Delta x, y, z) \\ \omega_P = \omega(x, y, z) \equiv 0 & ; \omega_A = \omega(x + \Delta x, y, z) \equiv 0 \end{array}$$



DILATAZIONI

$\overline{PA}$: fibra di lunghezza iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [\mu, \nu, 0]^T$ ($w = 0$)

$\overline{P^*A^*}$: fibra deformata di lunghezza l_1

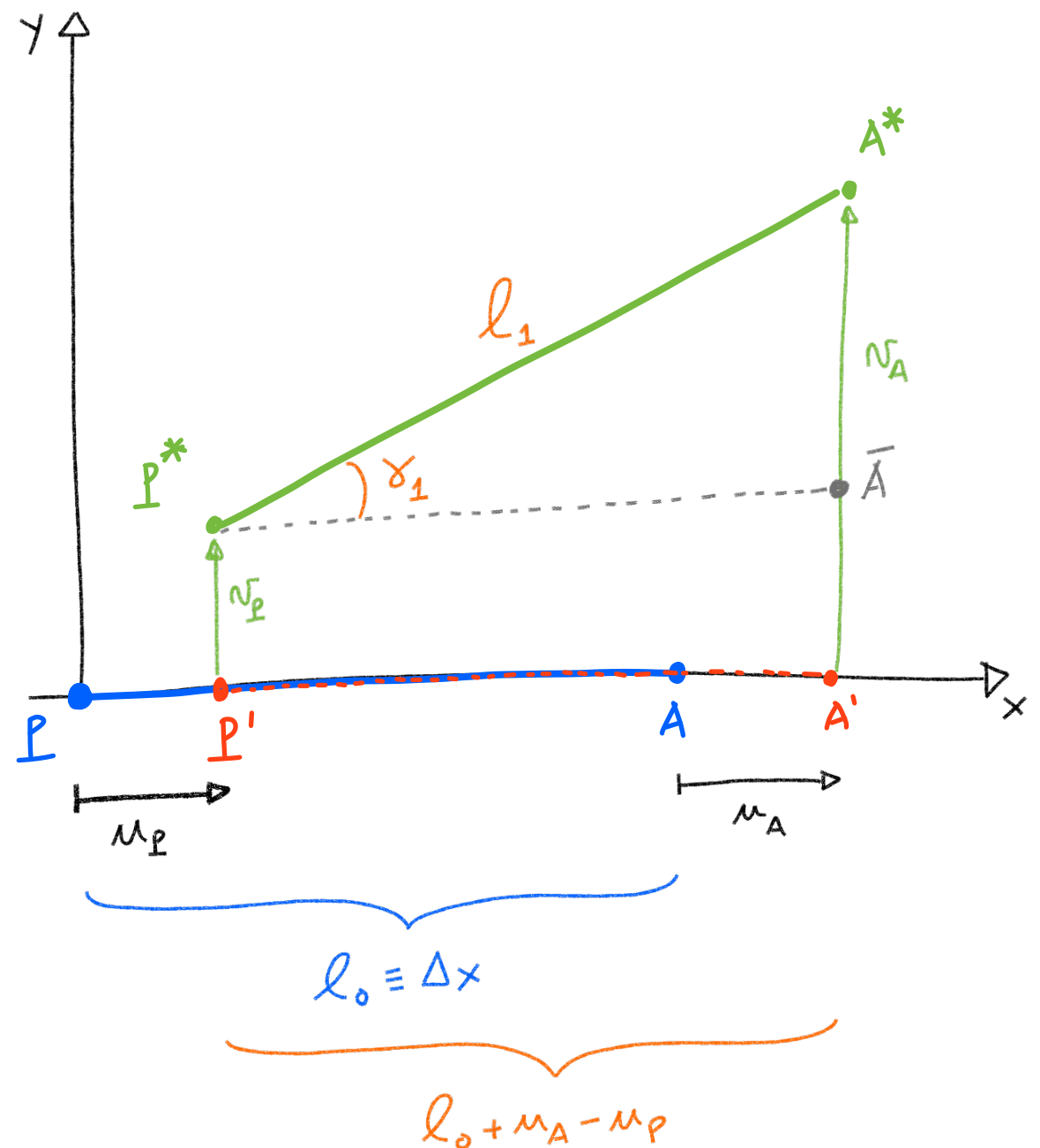
$P(x, y, z)$; $A(x + \Delta x, y, z)$

$\mu_P = \mu(x, y, z)$; $\mu_A = \mu(x + \Delta x, y, z)$

$\nu_P = \nu(x, y, z)$; $\nu_A = \nu(x + \Delta x, y, z)$

$w_P = w(x, y, z) \equiv 0$; $w_A = w(x + \Delta x, y, z) \equiv 0$

CALCOLO LUNGHEZZA DEFORMATA:



DILATAZIONI

$\overline{PA}$: fibra di lunghezza iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [\mu, \nu, 0]^T$ ($w = 0$)

$\overline{P^*A^*}$: fibra deformata di lunghezza l_1

* APPROX.: $\tan \delta_1 \approx 0$

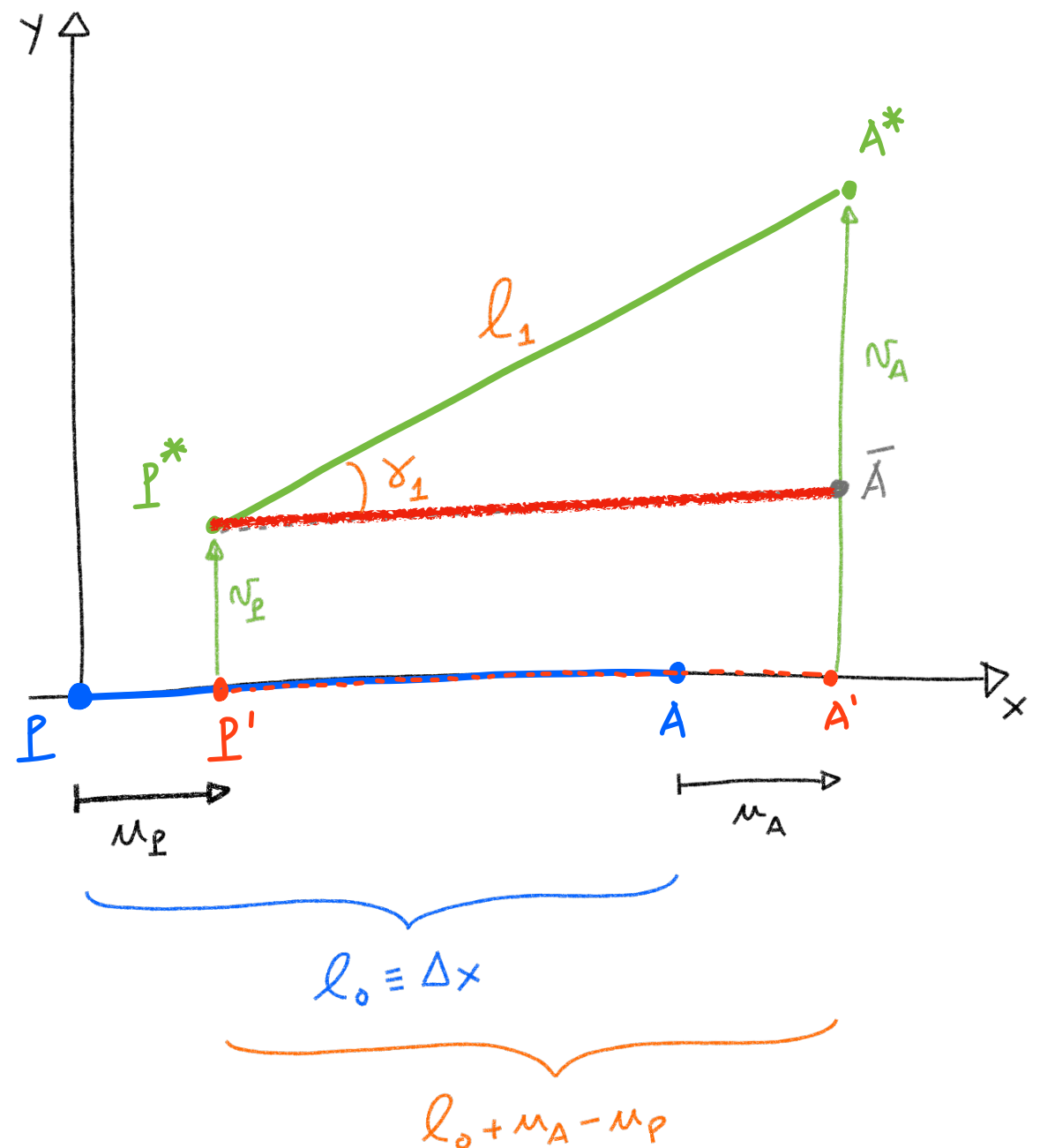
$P(x, y, z)$; $A(x + \Delta x, y, z)$

$\mu_P = \mu(x, y, z)$; $\mu_A = \mu(x + \Delta x, y, z)$

$\nu_P = \nu(x, y, z)$; $\nu_A = \nu(x + \Delta x, y, z)$

$w_P = w(x, y, z) \equiv 0$; $w_A = w(x + \Delta x, y, z) \equiv 0$

CALCOLO LUNGHEZZA DEFORMATA:



DILATAZIONI

$\overline{PA}$: fibra di lunghezza iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [\mu, \nu, 0]^T$ ($w = 0$)

$\overline{P^*A^*}$: fibra deformata di lunghezza l_1

* APPROX. : $\tan \gamma_1 \approx 0$

$P(x, y, z)$; $A(x + \Delta x, y, z)$

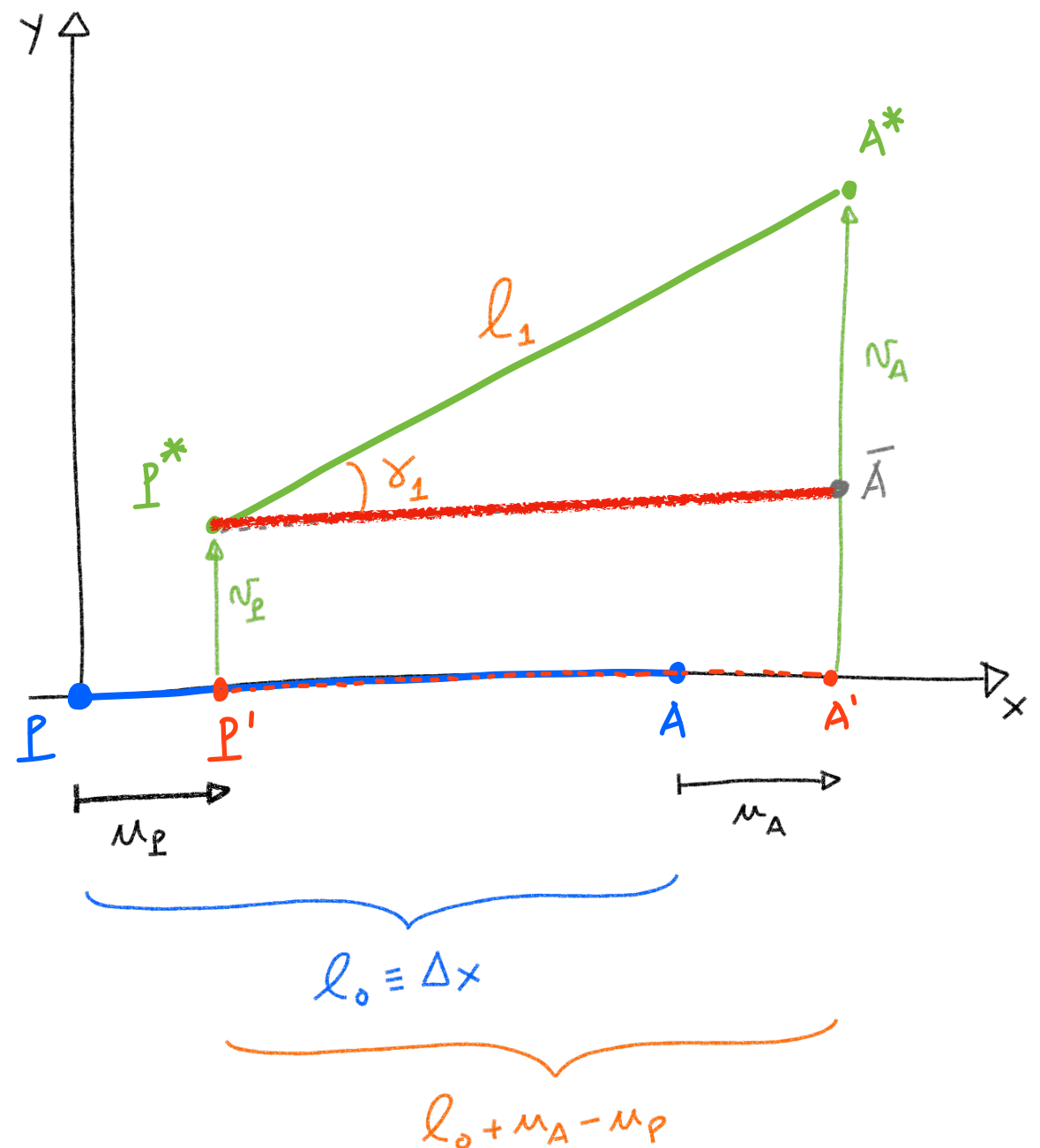
$\mu_P = \mu(x, y, z)$; $\mu_A = \mu(x + \Delta x, y, z)$

$\nu_P = \nu(x, y, z)$; $\nu_A = \nu(x + \Delta x, y, z)$

$w_P = w(x, y, z) \equiv 0$; $w_A = w(x + \Delta x, y, z) \equiv 0$

CALCOLO LUNGHEZZA DEFORMATA:

$$\begin{aligned} l_1 = \overline{P^*A^*} &\simeq \overline{P^*\bar{A}} = l_0 + \mu_A - \mu_P \\ &= l_0 + \mu(x + \Delta x, y, z) - \mu(x, y, z) \end{aligned}$$



DILATAZIONI

$\overline{PA}$: fibra di lunghezza iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [\mu, \nu, 0]^T$ ($w = 0$)

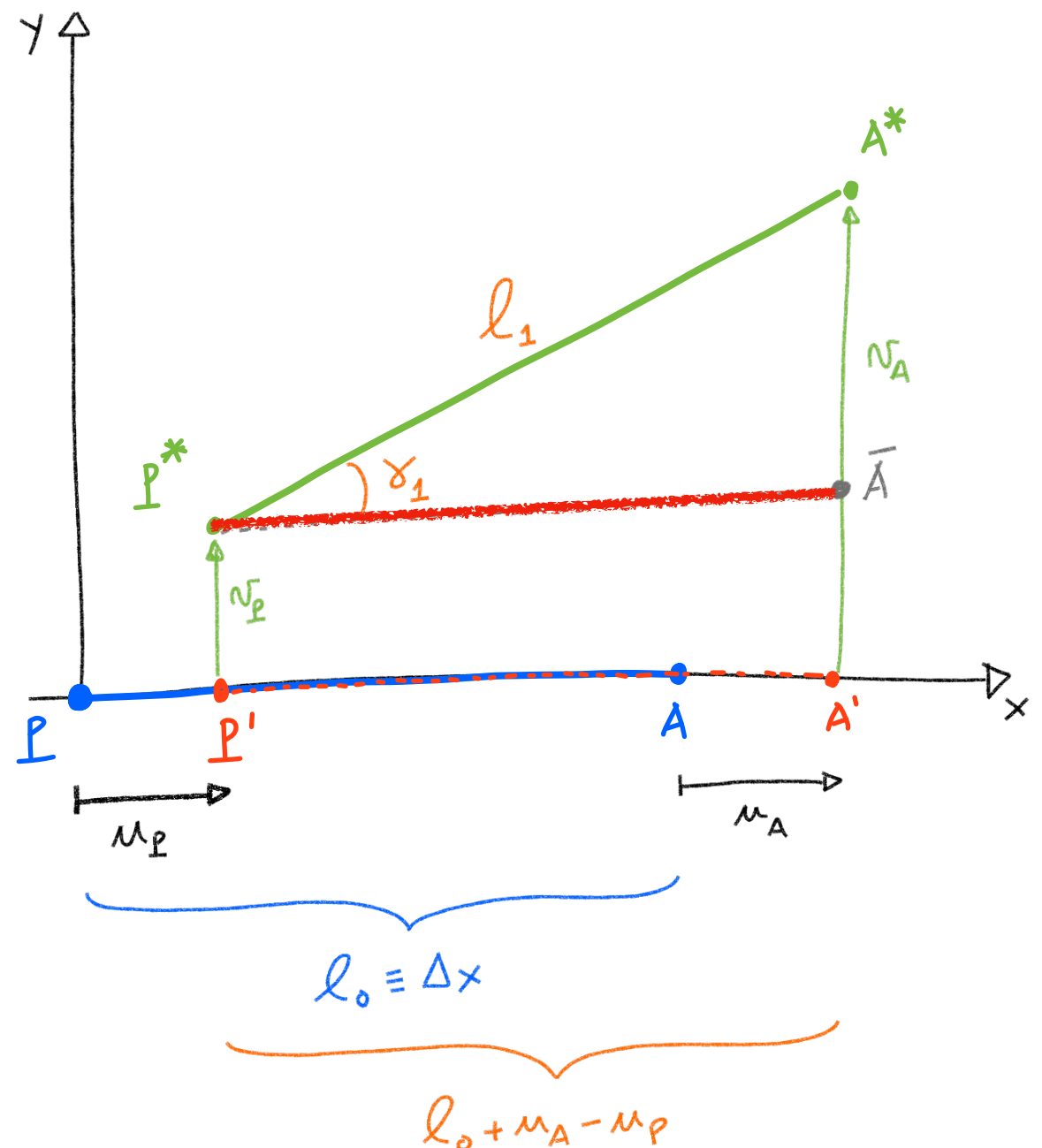
$\overline{P^*A^*}$: fibra deformata di lunghezza l_1

* APPROX.: $\tan \delta_1 \approx 0$

$$\begin{aligned} P(x, y, z) & ; A(x + \Delta x, y, z) \\ \mu_P = \mu(x, y, z) & ; \mu_A = \mu(x + \Delta x, y, z) \\ \nu_P = \nu(x, y, z) & ; \nu_A = \nu(x + \Delta x, y, z) \\ w_P = w(x, y, z) = 0 & ; w_A = w(x + \Delta x, y, z) = 0 \end{aligned}$$

CALCOLO LUNGHEZZA DEFORMATA:

$$\begin{aligned} l_1 = \overline{P^*A^*} & \simeq \overline{P^*\bar{A}} = l_0 + \mu_A - \mu_P \\ & = l_0 + \mu(x + \Delta x, y, z) - \mu(x, y, z) \end{aligned}$$



DILATAZIONE
ASSIALE

$$\varepsilon_x = \lim_{\Delta x \rightarrow 0} \frac{l_1 - l_0}{l_0} = \frac{\mu(x + \Delta x, y, z) - \mu(x, y, z)}{\Delta x} = \frac{\partial \mu}{\partial x}$$

DILATAZIONI

$\overline{PA}$: fibra di lunghezza iniziale $l_0 = \Delta x$

H_p : SPOSTAMENTO PIANO $\rightarrow \underline{u} = [\mu, \nu, 0]^T$ ($w = 0$)

$\overline{P^*A^*}$: fibra deformata di lunghezza l_1

* APPROX.: $\tan \delta_1 \approx 0$

$$\begin{aligned} P(x, y, z) & ; A(x + \Delta x, y, z) \\ \mu_P = \mu(x, y, z) & ; \mu_A = \mu(x + \Delta x, y, z) \\ \nu_P = \nu(x, y, z) & ; \nu_A = \nu(x + \Delta x, y, z) \\ w_P = w(x, y, z) = 0 & ; w_A = w(x + \Delta x, y, z) = 0 \end{aligned}$$

CALCOLO LUNGHEZZA DEFORMATA:

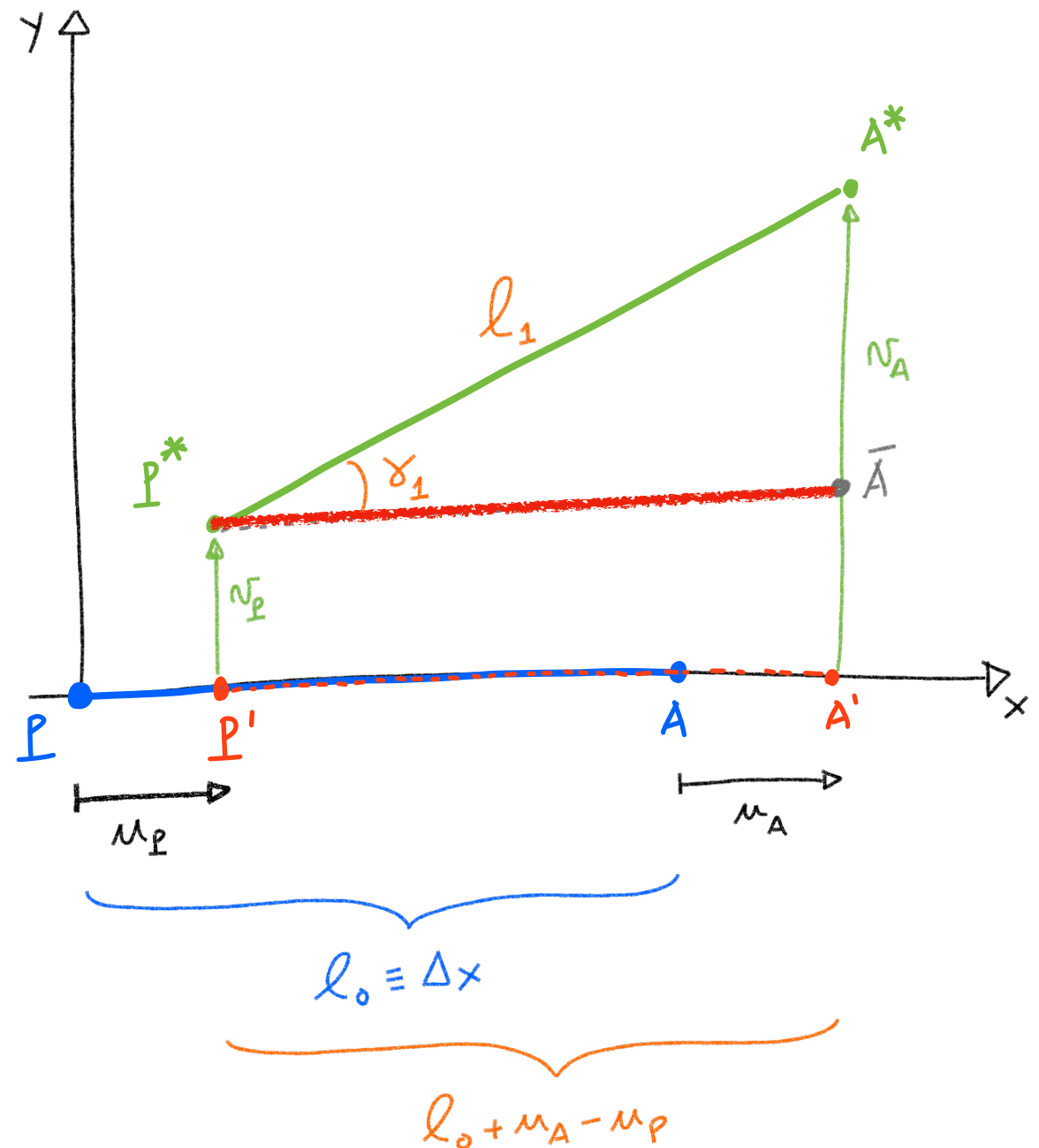
$$\begin{aligned} l_1 = \overline{P^*A^*} & \simeq \overline{P^*\bar{A}} = l_0 + \mu_A - \mu_P \\ & = l_0 + \mu(x + \Delta x, y, z) - \mu(x, y, z) \end{aligned}$$

DILATAZIONE
ASSIALE

$$\varepsilon_x = \lim_{\Delta x \rightarrow 0} \frac{l_1 - l_0}{l_0} = \frac{\mu(x + \Delta x, y, z) - \mu(x, y, z)}{\Delta x} = \frac{\partial \mu}{\partial x}$$

c.v.d.

ELEMENTI SULLA
DIAGONALE DI $\underline{\underline{\epsilon}}$



SCORRIMENTI ANGOLARI

SCORRIMENTI ANGOLARI

$\overline{PA}$: fibra Δx

$$\mu_P = \mu(x, y, z)$$

$$N_P = N(x, y, z)$$

$$\omega_P = \omega(x, y, z)$$

$\overline{PB}$: fibra Δy

$$\mu_A = \mu(x + \Delta x, y, z)$$

$$N_A = N(x + \Delta x, y, z)$$

$$\omega_P = \omega(x + \Delta x, y, z)$$

$$\mu_B = \mu(x, y + \Delta y, z)$$

$$N_B = N(x, y + \Delta y, z)$$

$$\omega_B = \omega(x, y + \Delta y, z)$$

SCORRIMENTI ANGOLARI

$\overline{PA}$: fibra Δx

$\overline{PB}$: fibra Δy

$$\mu_P = \mu(x, y, z)$$

$$\mu_A = \mu(x + \Delta x, y, z)$$

$$\mu_B = \mu(x, y + \Delta y, z)$$

$$\nu_P = \nu(x, y, z)$$

$$\nu_A = \nu(x + \Delta x, y, z)$$

$$\nu_B = \nu(x, y + \Delta y, z)$$

$$\omega_P = \omega(x, y, z)$$

$$\omega_P = \omega(x + \Delta x, y, z)$$

$$\omega_B = \omega(x, y + \Delta y, z)$$

ψ_{xy} : ANGOLO TRA LE FIBRE NELLA
CONFIGURAZIONE DEFORMATA

SCORRIMENTI ANGOLARI

$\overline{PA}$: fibra Δx

$\overline{PB}$: fibra Δy

$$\mu_P = \mu(x, y, z)$$

$$\mu_A = \mu(x + \Delta x, y, z)$$

$$\mu_B = \mu(x, y + \Delta y, z)$$

$$\nu_P = \nu(x, y, z)$$

$$\nu_A = \nu(x + \Delta x, y, z)$$

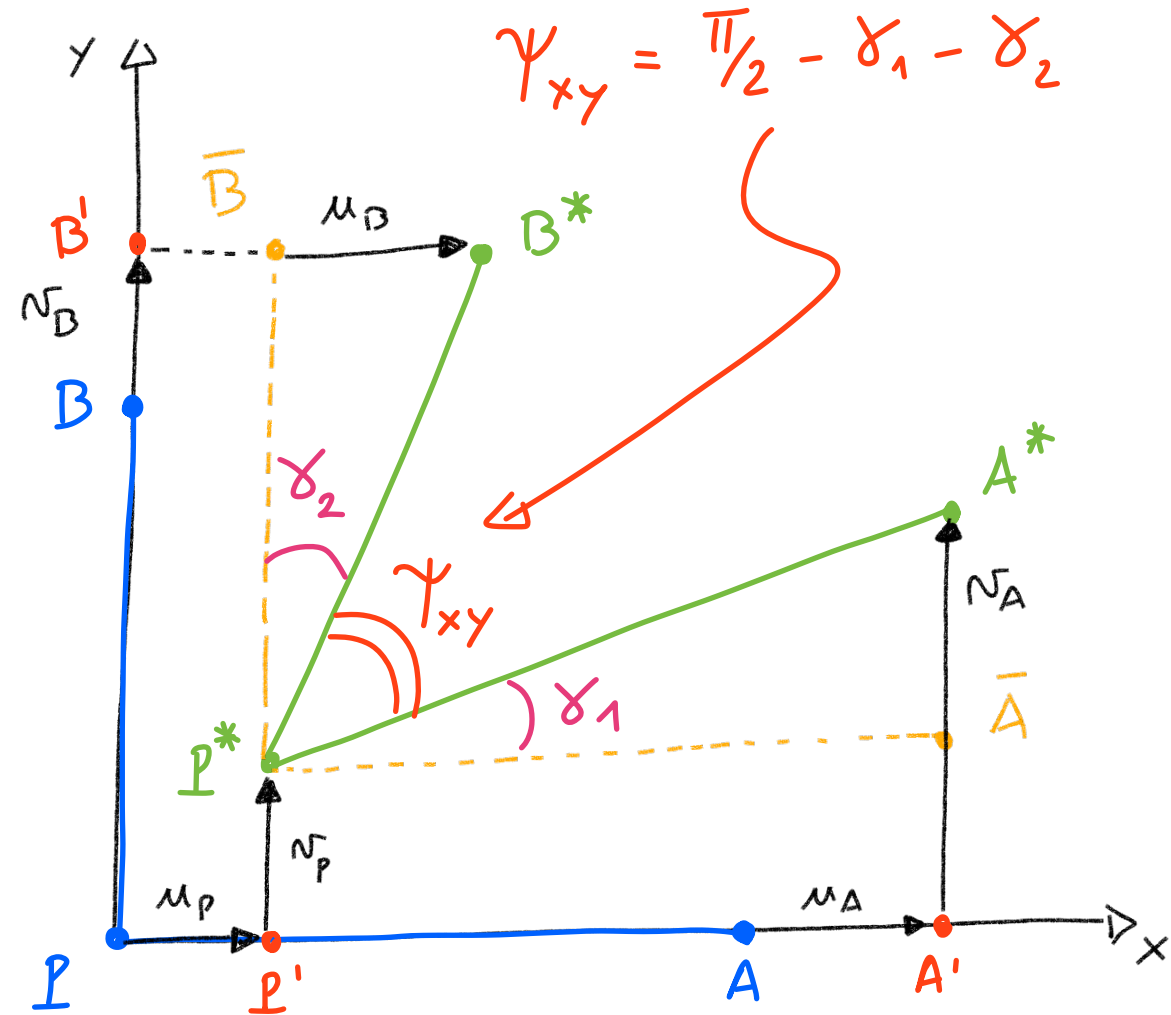
$$\nu_B = \nu(x, y + \Delta y, z)$$

$$\omega_P = \omega(x, y, z)$$

$$\omega_P = \omega(x + \Delta x, y, z)$$

$$\omega_B = \omega(x, y + \Delta y, z)$$

γ_{xy} : ANGOLO TRA LE FIBRE NELLA CONFIGURAZIONE DEFORMATA



SCORRIMENTI ANGOLARI

$\overline{PA}$: fibra Δx

$\overline{PB}$: fibra Δy

$$\mu_P = \mu(x, y, z)$$

$$\mu_A = \mu(x + \Delta x, y, z)$$

$$\mu_B = \mu(x, y + \Delta y, z)$$

$$\nu_P = \nu(x, y, z)$$

$$\nu_A = \nu(x + \Delta x, y, z)$$

$$\nu_B = \nu(x, y + \Delta y, z)$$

$$\omega_P = \omega(x, y, z)$$

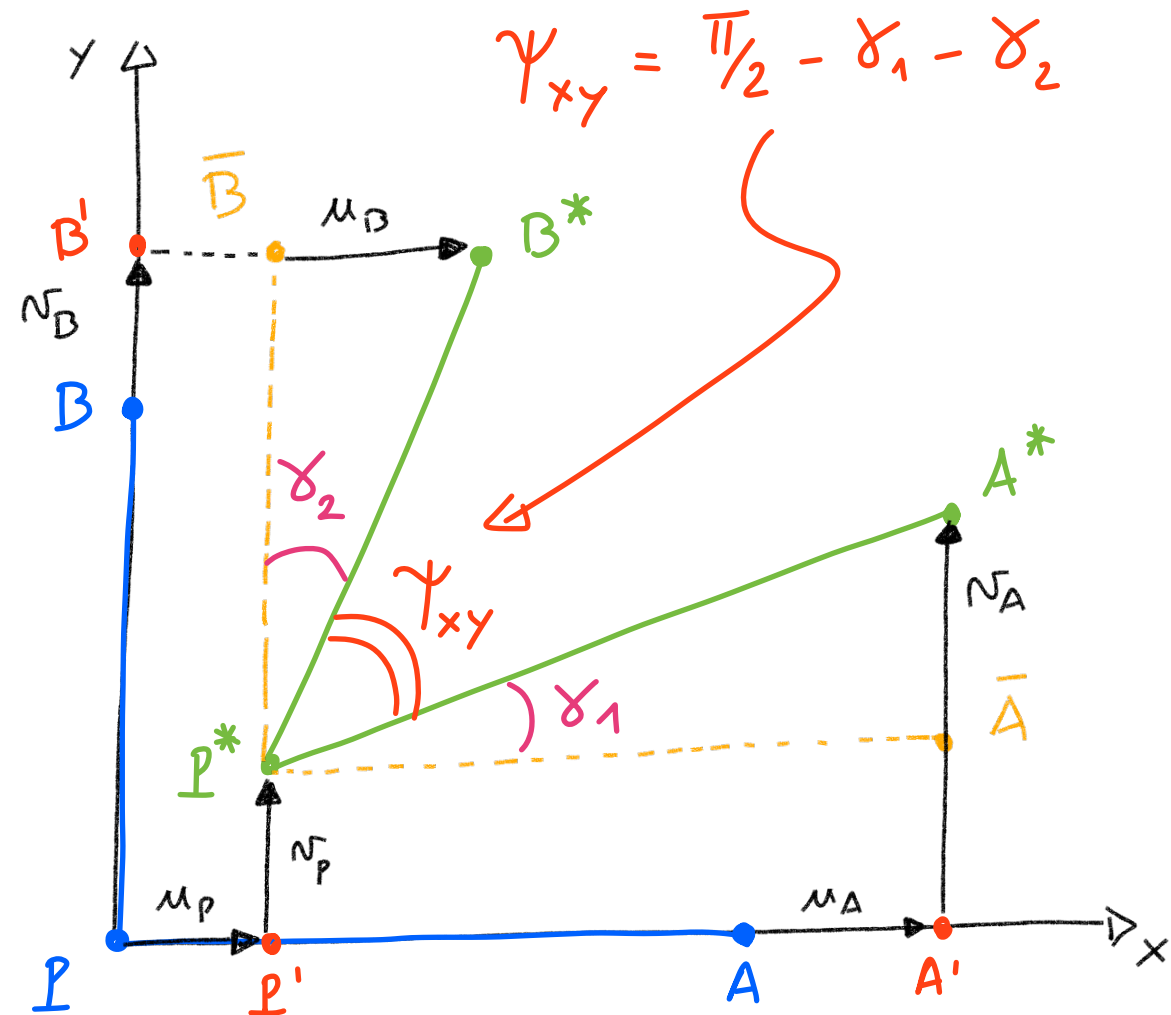
$$\omega_P = \omega(x + \Delta x, y, z)$$

$$\omega_B = \omega(x, y + \Delta y, z)$$

γ_{xy} : ANGOLO TRA LE FIBRE NELLA CONFIGURAZIONE DEFORMATA

$$\gamma_1 \approx \tan \gamma_1 = \frac{\overline{AA^*}}{\overline{P^*A}} \approx \frac{\overline{AA^*}}{\overline{PA}} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\nu(x + \Delta x, y, z) - \nu(x, y, z)}{\Delta x} = \frac{\partial \nu}{\partial x}$$



SCORRIMENTI ANGOLARI

$\overline{PA}$: fibra Δx

$\overline{PB}$: fibra Δy

$$\mu_P = \mu(x, y, z)$$

$$\mu_A = \mu(x + \Delta x, y, z)$$

$$\mu_B = \mu(x, y + \Delta y, z)$$

$$\nu_P = \nu(x, y, z)$$

$$\nu_A = \nu(x + \Delta x, y, z)$$

$$\nu_B = \nu(x, y + \Delta y, z)$$

$$\omega_P = \omega(x, y, z)$$

$$\omega_P = \omega(x + \Delta x, y, z)$$

$$\omega_B = \omega(x, y + \Delta y, z)$$

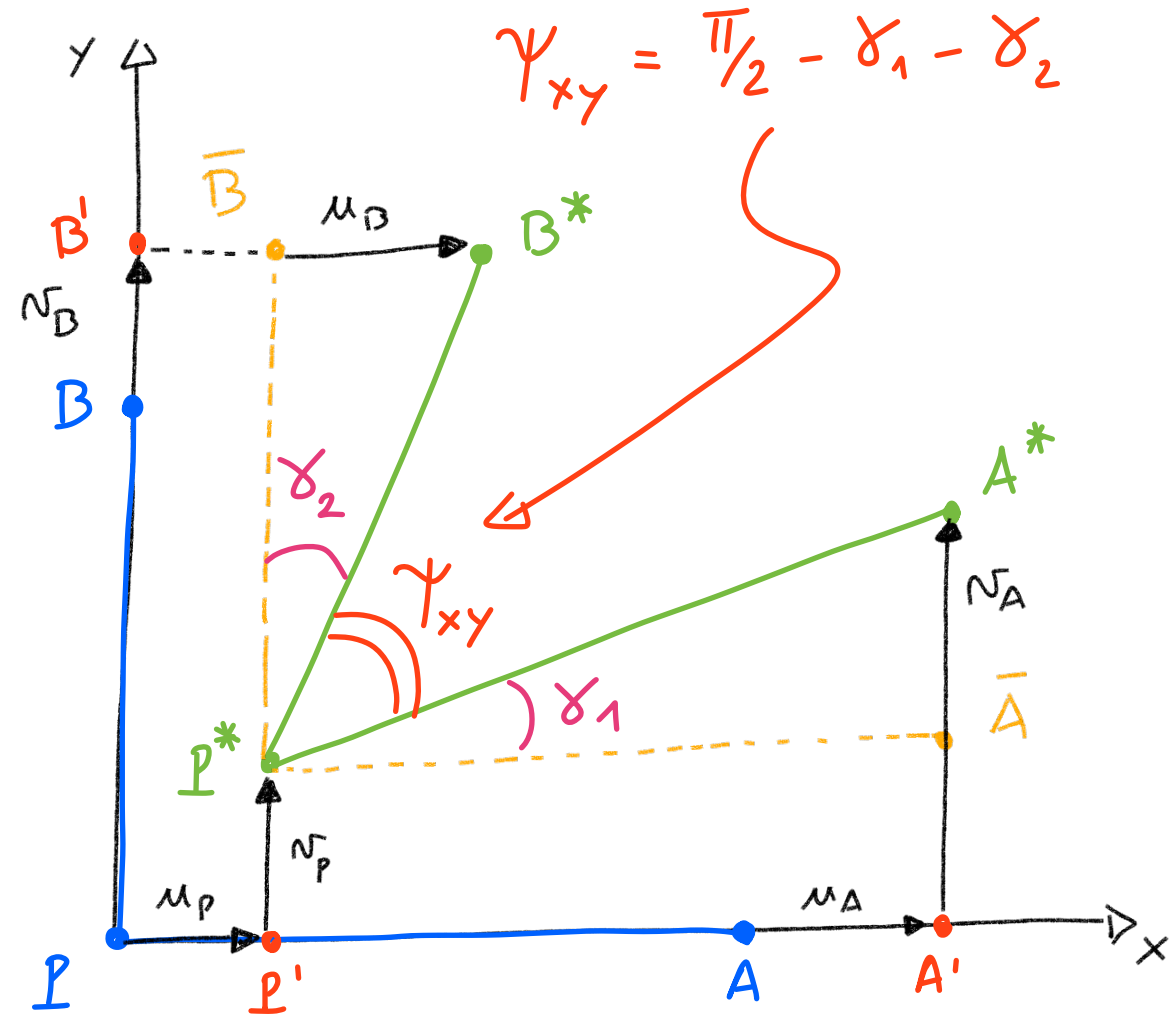
γ_{xy} : ANGOLO TRA LE FIBRE NELLA CONFIGURAZIONE DEFORMATA

$$\gamma_1 \approx \tan \gamma_1 = \frac{\overline{AA^*}}{\overline{P^*A}} \approx \frac{\overline{AA^*}}{\overline{PA}} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\nu(x + \Delta x, y, z) - \nu(x, y, z)}{\Delta x} = \frac{\partial \nu}{\partial x}$$

$$\gamma_2 \approx \tan \gamma_2 = \frac{\overline{BB^*}}{\overline{P^*B}} \approx \frac{\overline{BB^*}}{\overline{PB}} =$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\mu(x, y + \Delta y, z) - \mu(x, y, z)}{\Delta y} = \frac{\partial \mu}{\partial y}$$



SCORRIMENTI ANGOLARI

$\overline{PA}$: fibra Δx

$\overline{PB}$: fibra Δy

$$\mu_P = \mu(x, y, z)$$

$$\mu_A = \mu(x + \Delta x, y, z)$$

$$\mu_B = \mu(x, y + \Delta y, z)$$

$$\nu_P = \nu(x, y, z)$$

$$\nu_A = \nu(x + \Delta x, y, z)$$

$$\nu_B = \nu(x, y + \Delta y, z)$$

$$\omega_P = \omega(x, y, z)$$

$$\omega_P = \omega(x + \Delta x, y, z)$$

$$\omega_B = \omega(x, y + \Delta y, z)$$

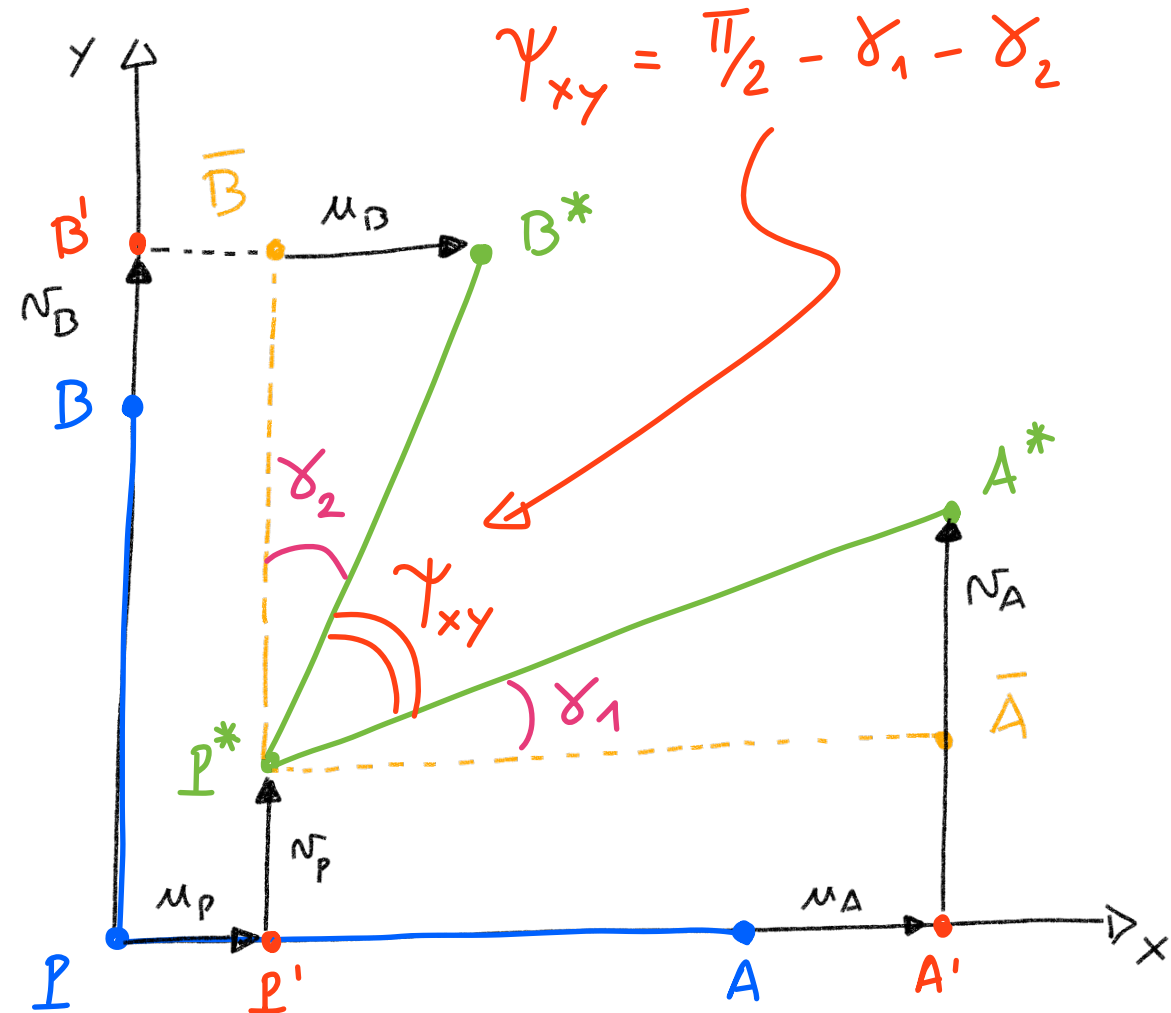
ψ_{xy} : ANGOLO TRA LE FIBRE NELLA CONFIGURAZIONE DEFORMATA

$$\delta_1 \approx \tan \delta_1 = \frac{\overline{AA^*}}{\overline{P^*A}} \approx \frac{\overline{AA^*}}{\overline{PA}} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\nu(x + \Delta x, y, z) - \nu(x, y, z)}{\Delta x} = \frac{\partial \nu}{\partial x}$$

$$\delta_2 \approx \tan \delta_2 = \frac{\overline{BB^*}}{\overline{P^*B}} \approx \frac{\overline{BB^*}}{\overline{PB}} =$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\mu(x, y + \Delta y, z) - \mu(x, y, z)}{\Delta y} = \frac{\partial \mu}{\partial y}$$



$$\psi_{xy} = \frac{\pi}{2} - \left(\frac{\partial \nu}{\partial x} + \frac{\partial \mu}{\partial y} \right) = \frac{\pi}{2} - \gamma_{xy}$$

SCORRIMENTI ANGOLARI

$\overline{PA}$: fibra Δx

$\overline{PB}$: fibra Δy

$$\mu_P = \mu(x, y, z)$$

$$\mu_A = \mu(x + \Delta x, y, z)$$

$$\mu_B = \mu(x, y + \Delta y, z)$$

$$\nu_P = \nu(x, y, z)$$

$$\nu_A = \nu(x + \Delta x, y, z)$$

$$\nu_B = \nu(x, y + \Delta y, z)$$

$$\omega_P = \omega(x, y, z)$$

$$\omega_P = \omega(x + \Delta x, y, z)$$

$$\omega_B = \omega(x, y + \Delta y, z)$$

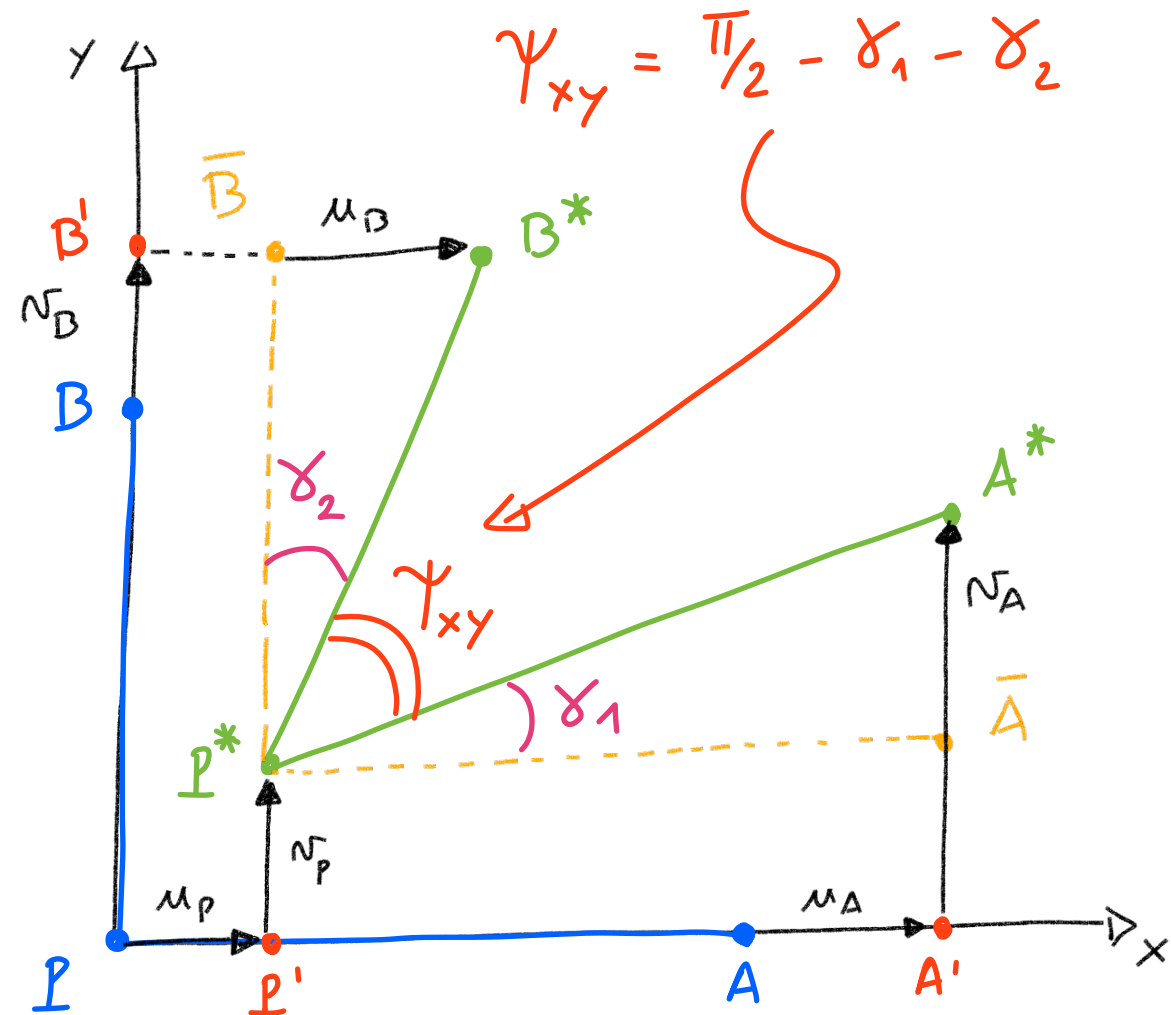
ψ_{xy} : ANGOLO TRA LE FIBRE NELLA CONFIGURAZIONE DEFORMATA

$$\delta_1 \approx \tan \delta_1 = \frac{\overline{AA^*}}{\overline{P^*A}} \approx \frac{\overline{AA^*}}{\overline{PA}} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\nu(x + \Delta x, y, z) - \nu(x, y, z)}{\Delta x} = \frac{\partial \nu}{\partial x}$$

$$\delta_2 \approx \tan \delta_2 = \frac{\overline{BB^*}}{\overline{P^*B}} \approx \frac{\overline{BB^*}}{\overline{PB}} =$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\mu(x, y + \Delta y, z) - \mu(x, y, z)}{\Delta y} = \frac{\partial \mu}{\partial y}$$



$$\left\{ \begin{array}{l} \psi_{xz} = \pi/2 - \delta_{xz} \\ \psi_{yz} = \pi/2 - \delta_{yz} \end{array} \right\}$$

$$\psi_{xy} = \frac{\pi}{2} - \left(\frac{\partial \nu}{\partial x} + \frac{\partial \mu}{\partial y} \right) = \frac{\pi}{2} - \gamma_{xy}$$

SCORRIMENTI ANGOLARI

$\overline{PA}$: fibra Δx

$\overline{PB}$: fibra Δy

$$\mu_P = \mu(x, y, z)$$

$$\mu_A = \mu(x + \Delta x, y, z)$$

$$\mu_B = \mu(x, y + \Delta y, z)$$

$$\nu_P = \nu(x, y, z)$$

$$\nu_A = \nu(x + \Delta x, y, z)$$

$$\nu_B = \nu(x, y + \Delta y, z)$$

$$\omega_P = \omega(x, y, z)$$

$$\omega_P = \omega(x + \Delta x, y, z)$$

$$\omega_B = \omega(x, y + \Delta y, z)$$

ψ_{xy} : ANGOLO TRA LE FIBRE NELLA CONFIGURAZIONE DEFORMATA

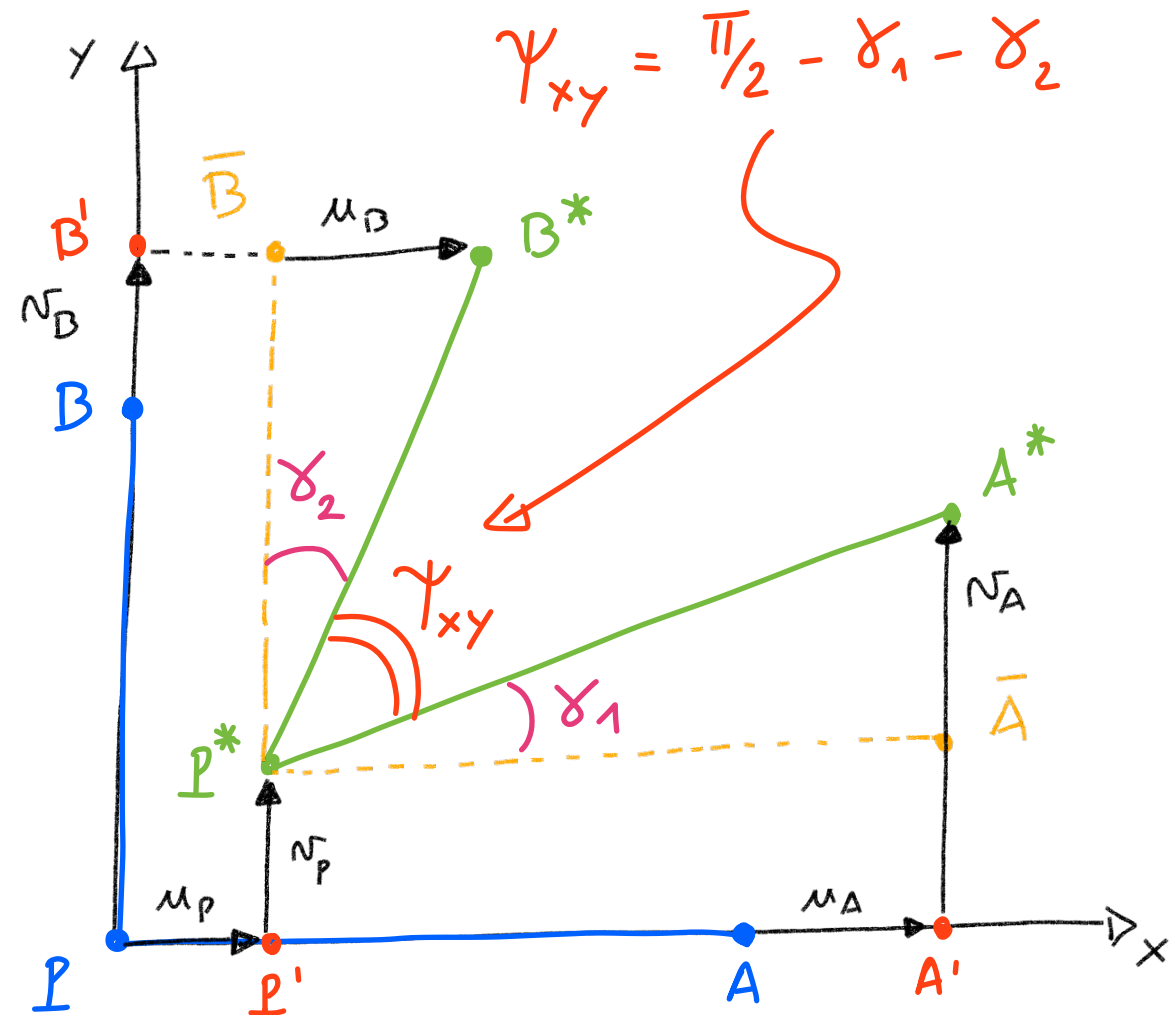
$$\delta_1 \approx \tan \delta_1 = \frac{\overline{AA^*}}{\overline{P^*A}} \approx \frac{\overline{AA^*}}{\overline{PA}} =$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\nu(x + \Delta x, y, z) - \nu(x, y, z)}{\Delta x} = \frac{\partial \nu}{\partial x}$$

$$\delta_2 \approx \tan \delta_2 = \frac{\overline{BB^*}}{\overline{P^*B}} \approx \frac{\overline{BB^*}}{\overline{PB}} =$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\mu(x, y + \Delta y, z) - \mu(x, y, z)}{\Delta y} = \frac{\partial \mu}{\partial y}$$

$$\psi_{xy} = \frac{\pi}{2} - \left(\frac{\partial \nu}{\partial x} + \frac{\partial \mu}{\partial y} \right) = \frac{\pi}{2} - \gamma_{xy}$$



$$\left\{ \begin{array}{l} \psi_{xz} = \pi/2 - \gamma_{xz} \\ \psi_{yz} = \pi/2 - \gamma_{yz} \end{array} \right\}$$

ELEMENTI FUORI DIAGONALE DI $\underline{\underline{E}}$ RAPPRESENTANO GLI SCORRIMENTI ANGOLARI TRA LE FIBRE

PROCESSO DEFORMATIVO

PROCESSO DEFORMATIVO

- 1) TRASLAZIONE RIGIDA : $\underline{u}_p$
- 2) ROTAZIONE RIGIDA : $\underline{\omega}$
- 3) DEFORMAZIONE PURA : $\underline{E}(\epsilon, \gamma)$
(TRASFORMAZIONE AFFINE)

PROCESSO DEFORMATIVO

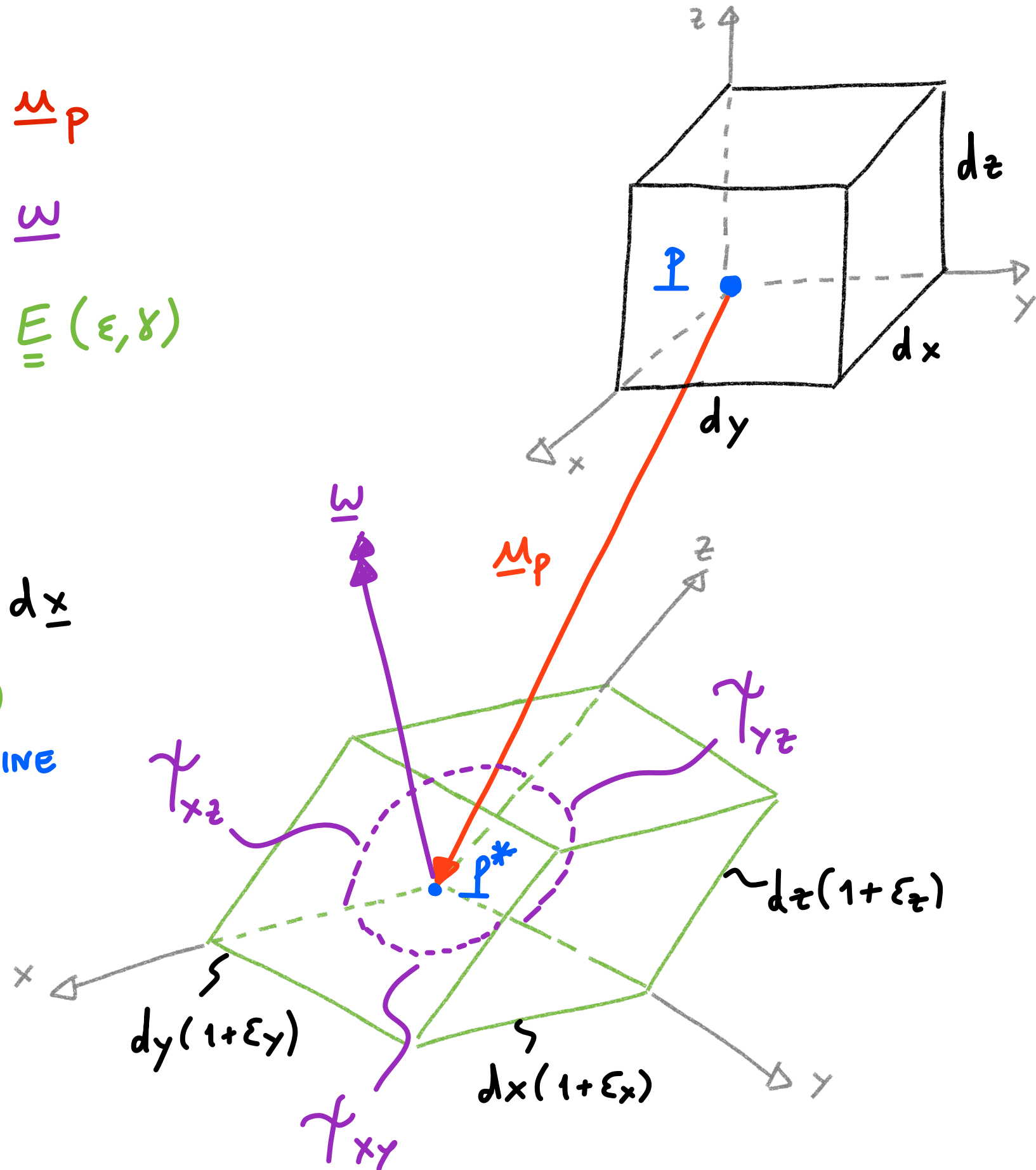
- 1) TRASLAZIONE RIGIDA : $\underline{u}_p$
- 2) ROTAZIONE RIGIDA : $\underline{\omega}$
- 3) DEFORMAZIONE PURA : $\underline{E}(\epsilon, \gamma)$
(TRASFORMAZIONE AFFINE)

$$\underline{u}_Q = \underbrace{\underline{u}_p}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega} d\underline{x}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{E} d\underline{x}}_{\textcircled{3} \text{ AFFINE}}$$

PROCESSO DEFORMATIVO

- 1) TRASLAZIONE RIGIDA : $\underline{u}_p$
- 2) ROTAZIONE RIGIDA : $\underline{\omega}$
- 3) DEFORMAZIONE PURA : $\underline{\epsilon}(\epsilon, \gamma)$
(TRASFORMAZIONE AFFINE)

$$\underline{u}_Q = \underbrace{\underline{u}_p}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega} d\underline{x}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{\epsilon} d\underline{x}}_{\textcircled{3} \text{ AFFINE}}$$



PROCESSO DEFORMATIVO

- 1) TRASLAZIONE RIGIDA : $\underline{u}_p$
- 2) ROTAZIONE RIGIDA : $\underline{\omega}$
- 3) DEFORMAZIONE PURA : $\underline{\epsilon}(\epsilon, \gamma)$
(TRASFORMAZIONE AFFINE)

$$\underline{u}_Q = \underline{u}_p + \underline{\Omega} d\underline{x} + \underline{\epsilon} d\underline{x}$$

①
RIGIDA

②
RIGIDA

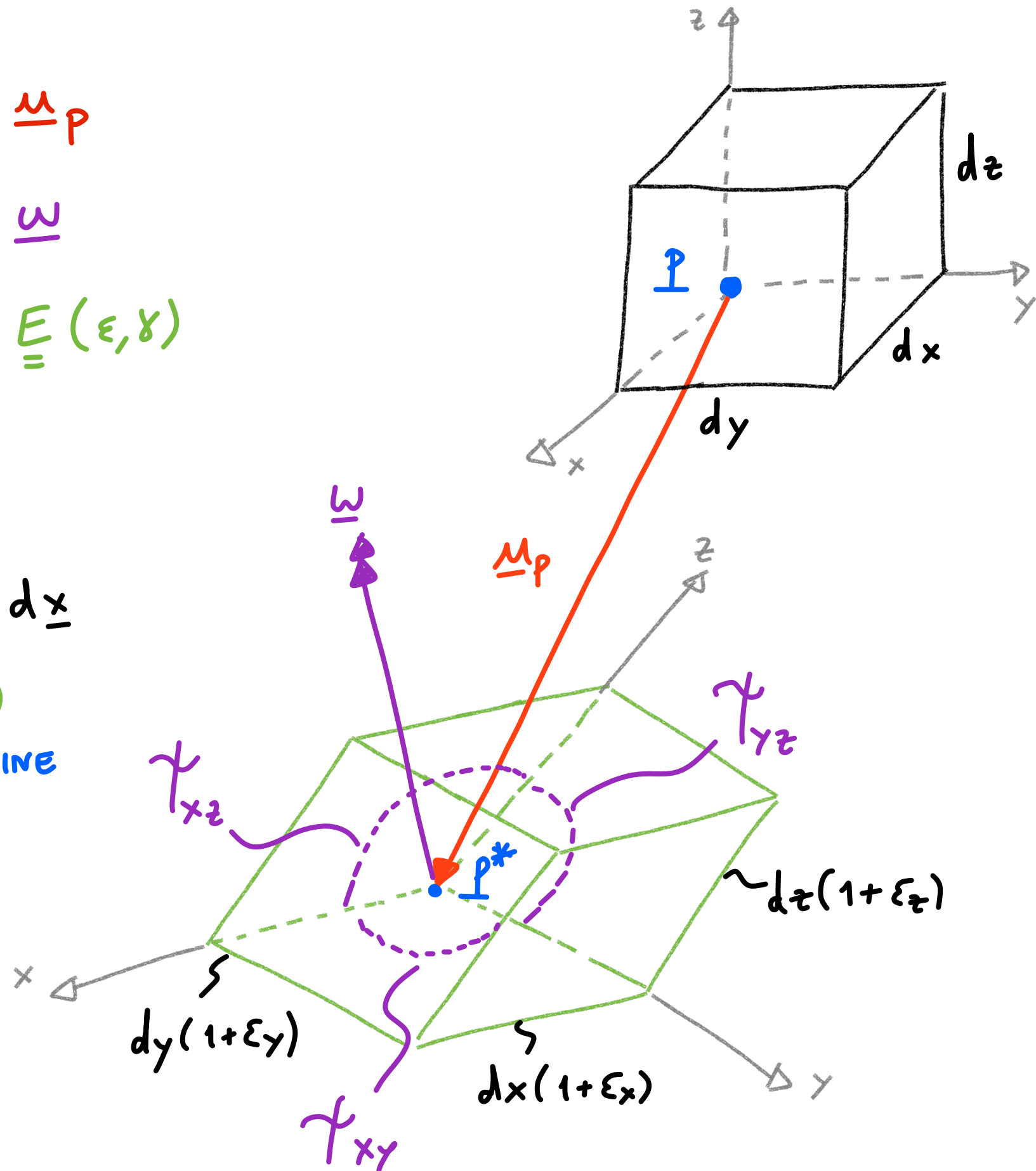
③
AFFINE

$$\begin{aligned} dx &\rightarrow dx(1+\epsilon_x) \\ dy &\rightarrow dy(1+\epsilon_y) \\ dz &\rightarrow dz(1+\epsilon_z) \end{aligned}$$

$$\gamma_{xy} = \pi/2 - \gamma_{xy}$$

$$\gamma_{xz} = \pi/2 - \gamma_{xz}$$

$$\gamma_{yz} = \pi/2 - \gamma_{yz}$$



DILATAZIONE CUBICA

DILATAZIONE CUBICA

$$V_0 = dx dy dz$$

DILATAZIONE CUBICA

$$V_0 = dx dy dz$$

$$\begin{aligned} dx &\rightarrow dx(1+\varepsilon_x) \\ dy &\rightarrow dy(1+\varepsilon_y) \\ dz &\rightarrow dz(1+\varepsilon_z) \end{aligned}$$

DILATAZIONE CUBICA

$$V_0 = dx dy dz$$

$$V_f = dx(1 + \varepsilon_x) \cdot dy(1 + \varepsilon_y) \cdot dz(1 + \varepsilon_z)$$

$$dx \rightarrow dx(1 + \varepsilon_x)$$

$$dy \rightarrow dy(1 + \varepsilon_y)$$

$$dz \rightarrow dz(1 + \varepsilon_z)$$

DILATAZIONE CUBICA

$$V_0 = dx dy dz$$

$$V_f = dx(1+\varepsilon_x) \cdot dy(1+\varepsilon_y) \cdot dz(1+\varepsilon_z)$$

$$= dx dy dz (1 + \varepsilon_x + \varepsilon_y + \varepsilon_z + \varepsilon_x \varepsilon_y + \varepsilon_x \varepsilon_z + \varepsilon_y \varepsilon_z + \varepsilon_x \varepsilon_y \varepsilon_z)$$

$$\begin{aligned} dx &\rightarrow dx(1+\varepsilon_x) \\ dy &\rightarrow dy(1+\varepsilon_y) \\ dz &\rightarrow dz(1+\varepsilon_z) \end{aligned}$$

DILATAZIONE CUBICA

$$V_0 = dx dy dz$$

$$\begin{aligned} dx &\rightarrow dx(1+\varepsilon_x) \\ dy &\rightarrow dy(1+\varepsilon_y) \\ dz &\rightarrow dz(1+\varepsilon_z) \end{aligned}$$

$$V_f = dx(1+\varepsilon_x) \cdot dy(1+\varepsilon_y) \cdot dz(1+\varepsilon_z)$$

$$= dx dy dz (1 + \varepsilon_x + \varepsilon_y + \varepsilon_z + \varepsilon_x \varepsilon_y + \varepsilon_x \varepsilon_z + \varepsilon_y \varepsilon_z + \varepsilon_x \varepsilon_y \varepsilon_z)$$

$$\simeq dx dy dz (1 + \varepsilon_x + \varepsilon_y + \varepsilon_z)$$

DILATAZIONE CUBICA

$$V_0 = dx dy dz$$

$$\begin{aligned} dx &\rightarrow dx(1+\varepsilon_x) \\ dy &\rightarrow dy(1+\varepsilon_y) \\ dz &\rightarrow dz(1+\varepsilon_z) \end{aligned}$$

$$V_f = dx(1+\varepsilon_x) \cdot dy(1+\varepsilon_y) \cdot dz(1+\varepsilon_z)$$

$$= dx dy dz (1 + \varepsilon_x + \varepsilon_y + \varepsilon_z + \varepsilon_x \varepsilon_y + \varepsilon_x \varepsilon_z + \varepsilon_y \varepsilon_z + \varepsilon_x \varepsilon_y \varepsilon_z)$$

$$\simeq dx dy dz (1 + \varepsilon_x + \varepsilon_y + \varepsilon_z)$$

$$= V_0 + (\varepsilon_x + \varepsilon_y + \varepsilon_z) V_0$$

DILATAZIONE CUBICA

$$V_0 = dx dy dz$$

$$\begin{aligned} dx &\rightarrow dx(1+\varepsilon_x) \\ dy &\rightarrow dy(1+\varepsilon_y) \\ dz &\rightarrow dz(1+\varepsilon_z) \end{aligned}$$

$$\begin{aligned} V_f &= dx(1+\varepsilon_x) \cdot dy(1+\varepsilon_y) \cdot dz(1+\varepsilon_z) \\ &= dx dy dz (1 + \varepsilon_x + \varepsilon_y + \varepsilon_z + \varepsilon_x \varepsilon_y + \varepsilon_x \varepsilon_z + \varepsilon_y \varepsilon_z + \varepsilon_x \varepsilon_y \varepsilon_z) \\ &\simeq dx dy dz (1 + \varepsilon_x + \varepsilon_y + \varepsilon_z) \\ &= V_0 + (\varepsilon_x + \varepsilon_y + \varepsilon_z) V_0 \end{aligned}$$

$$\rightarrow \frac{V_f - V_0}{V_0} = \varepsilon_x + \varepsilon_y + \varepsilon_z = \varepsilon_v = \text{tr}(\underline{\underline{E}}) = \sum_{i,j} E_{ij} = E_{ii}$$

VARIAZIONE DI VOLUME NELL'INTORNO DEL PUNTO

FORMULA DI CAUCHY PER LA DEFORMAZIONE

FORMULA DI CAUCHY PER LA DEFORMAZIONE

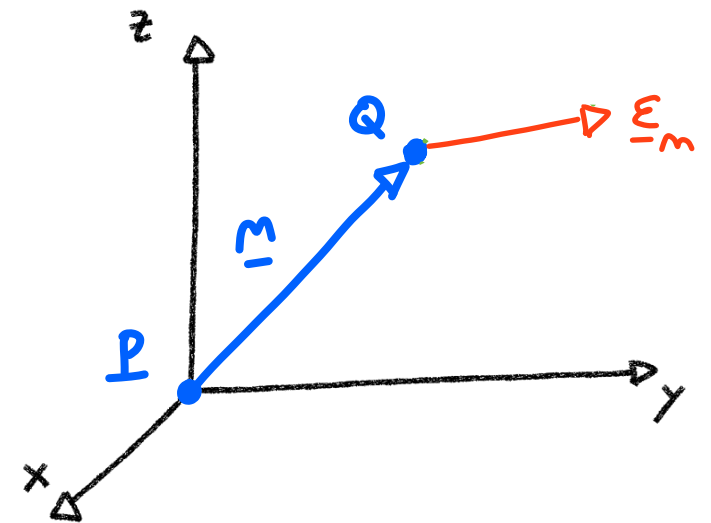
- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
- $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$

FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
- $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
$$\underline{m}_Q = \underbrace{\underline{m}_P}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega} d\underline{x}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{E} d\underline{x}}_{\textcircled{3} \text{ AFFINE}}$$

FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{u}_Q = \underbrace{\underline{u}_P}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega}}_{\textcircled{2} \text{ RIGIDA}} d\underline{x} + \underbrace{\underline{E}}_{\textcircled{3} \text{ AFFINE}} d\underline{x}$$

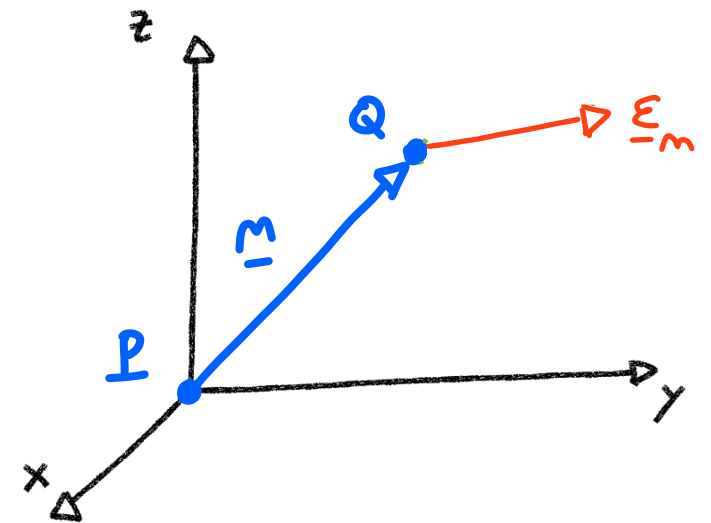


FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{u}_Q = \underbrace{\underline{u}_P}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega} d\underline{x}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{E} d\underline{x}}_{\textcircled{3} \text{ AFFINE}}$$

SPOSTAMENTO DI Q:

$$\underline{u}_Q = \underline{u}_P + \underline{E} \underline{m}_\delta + \underline{\Omega} \underline{m}_\delta = \underline{u}_P + \delta \underline{E} \underline{m} + \delta \underline{\Omega} \underline{m}$$



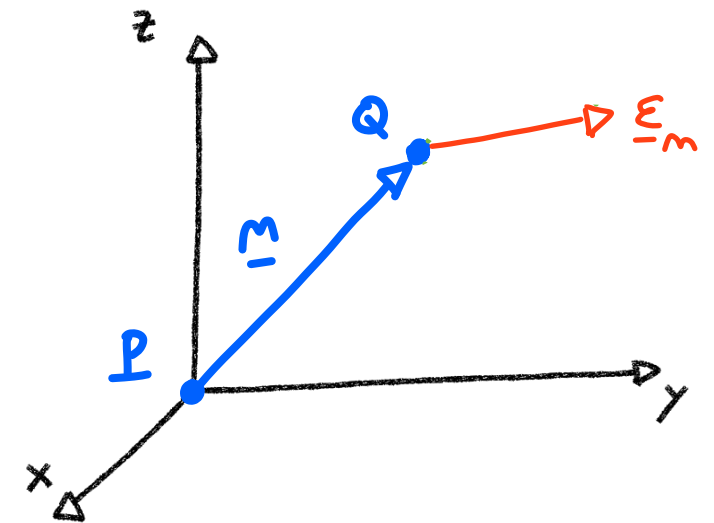
FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{u}_Q = \underline{u}_P + \underset{\textcircled{1} \text{ RIGIDA}}{\underline{\Omega}} d\underline{x} + \underset{\textcircled{2} \text{ RIGIDA}}{\underline{\Omega}} d\underline{x} + \underset{\textcircled{3} \text{ AFFINE}}{\underline{E}} d\underline{x}$$

SPOSTAMENTO DI Q:

$$\underline{u}_Q = \underline{u}_P + \underline{E} \underline{m}_\delta + \underline{\Omega} \underline{m}_\delta = \underline{u}_P + \delta \underline{E} \underline{m} + \cancel{\underline{\Omega} \underline{m}}$$

DEFORMAZIONE PURA : $\delta \underline{E} \underline{m} \mapsto$



FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{u}_Q = \underbrace{\underline{u}_P}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega} d\underline{x}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{E} d\underline{x}}_{\textcircled{3} \text{ AFFINE}}$$

SPOSTAMENTO DI Q:

$$\underline{u}_Q = \underline{u}_P + \underline{E} \underline{m}_\delta + \underline{\Omega} \underline{m}_\delta = \underline{u}_P + \delta \underline{E} \underline{m} + \cancel{\delta \underline{\Omega} \underline{m}}$$

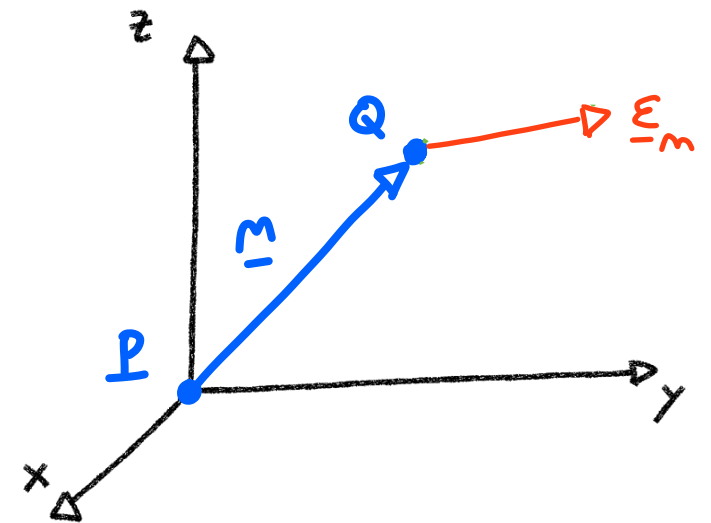
DEFORMAZIONE
PURA

$\delta \underline{E} \underline{m} \mapsto$

$$\underline{\varepsilon}_m = \underline{E} \underline{m}$$

VETTORE
DELLA
DEFORMAZIONE

FORMULA DI
CAUCHY PER LA
DEFORMAZIONE
(qualunque δ)



FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{u}_Q = \underbrace{\underline{u}_P}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega} d\underline{x}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{E} d\underline{x}}_{\textcircled{3} \text{ AFFINE}}$$

SPOSTAMENTO DI Q:

$$\underline{u}_Q = \underline{u}_P + \underline{E} \underline{m}_\delta + \underline{\Omega} \underline{m}_\delta = \underline{u}_P + \delta \underline{E} \underline{m} + \cancel{\underline{\Omega} \underline{m}}$$

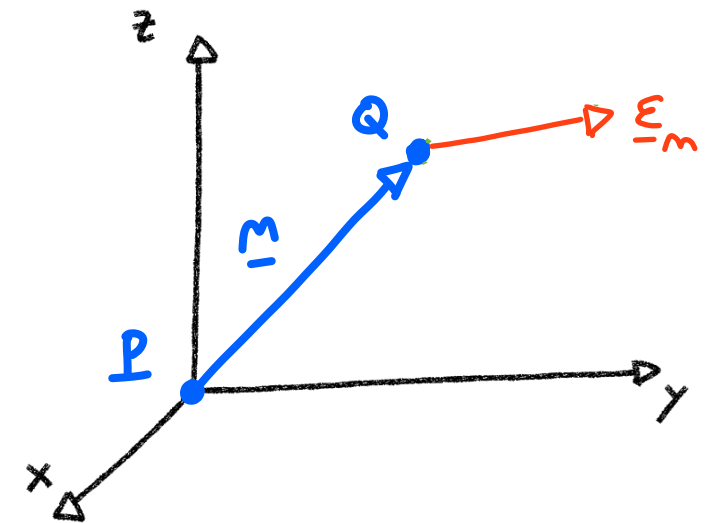
DEFORMAZIONE
PURA

$$\delta \underline{E} \underline{m} \mapsto$$

$$\underline{\varepsilon}_m = \underline{E} \underline{m}$$

VETTORE
DELLA
DEFORMAZIONE

FORMULA DI
CAUCHY PER LA
DEFORMAZIONE
(qualunque δ)



CALCOLO DELLE COMPONENTI

$$\underline{\varepsilon}_m = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ 1/2 \gamma_{xy} & \varepsilon_y & 1/2 \gamma_{yz} \\ 1/2 \gamma_{xz} & 1/2 \gamma_{yz} & \varepsilon_z \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \alpha \varepsilon_x + 1/2 \gamma_{xy} \beta + 1/2 \gamma_{xz} \gamma \\ 1/2 \gamma_{xy} \alpha + \beta \varepsilon_y + 1/2 \gamma_{yz} \gamma \\ 1/2 \gamma_{xz} \alpha + 1/2 \gamma_{yz} \beta + \gamma \varepsilon_z \end{bmatrix} \equiv \begin{bmatrix} \varepsilon_{mx} \\ \varepsilon_{my} \\ \varepsilon_{mz} \end{bmatrix}$$

FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{u}_Q = \underbrace{\underline{u}_P}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega} d\underline{x}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{E} d\underline{x}}_{\textcircled{3} \text{ AFFINE}}$$

SPOSTAMENTO DI Q:

$$\underline{u}_Q = \underline{u}_P + \underline{E} \underline{m}_\delta + \underline{\Omega} \underline{m}_\delta = \underline{u}_P + \delta \underline{E} \underline{m} + \cancel{\underline{\Omega} \underline{m}}$$

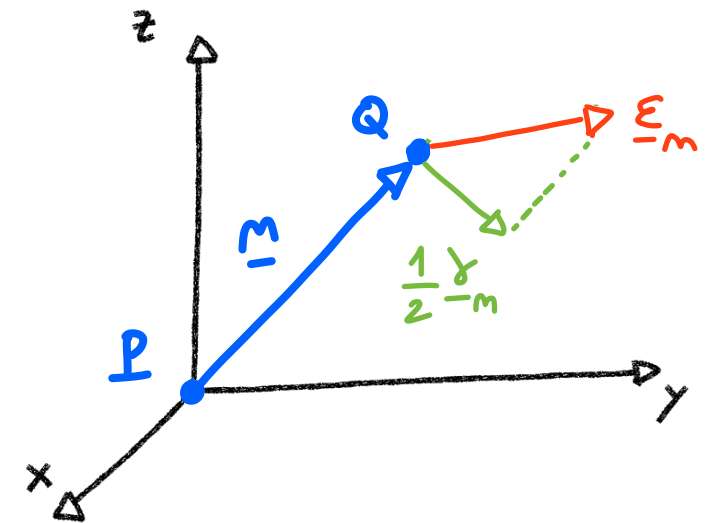
DEFORMAZIONE
PURA

$$\delta \underline{E} \underline{m} \mapsto$$

$$\underline{\varepsilon}_m = \underline{E} \underline{m}$$

VETTORE
DELLA
DEFORMAZIONE

FORMULA DI
CAUCHY PER LA
DEFORMAZIONE
(qualunque δ)



CALCOLO DELLE COMPONENTI

$$\underline{\varepsilon}_m = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ 1/2 \gamma_{xy} & \varepsilon_y & 1/2 \gamma_{yz} \\ 1/2 \gamma_{xz} & 1/2 \gamma_{yz} & \varepsilon_z \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \alpha \varepsilon_x + 1/2 \gamma_{xy} \beta + 1/2 \gamma_{xz} \gamma \\ 1/2 \gamma_{xy} \alpha + \beta \varepsilon_y + 1/2 \gamma_{yz} \gamma \\ 1/2 \gamma_{xz} \alpha + 1/2 \gamma_{yz} \beta + \gamma \varepsilon_z \end{bmatrix} \equiv \begin{bmatrix} \varepsilon_{mx} \\ \varepsilon_{my} \\ \varepsilon_{mz} \end{bmatrix}$$

FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\overline{PQ}$: FIBRA GENERICA $\equiv \underline{m}_\delta = \delta \underline{m} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{u}_Q = \underbrace{\underline{u}_P}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\Omega} d\underline{x}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{E} d\underline{x}}_{\textcircled{3} \text{ AFFINE}}$$

SPOSTAMENTO DI Q:

$$\underline{u}_Q = \underline{u}_P + \underline{E} \underline{m}_\delta + \underline{\Omega} \underline{m}_\delta = \underline{u}_P + \delta \underline{E} \underline{m} + \cancel{\underline{\Omega} \underline{m}}$$

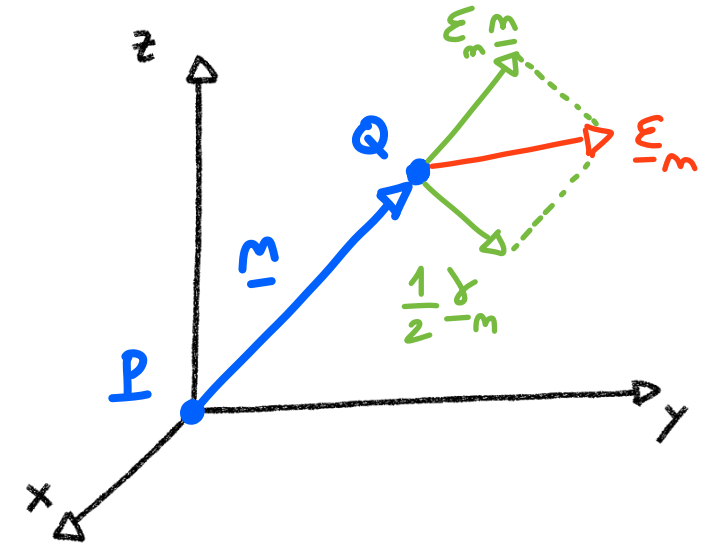
DEFORMAZIONE
PURA

$$\delta \underline{E} \underline{m} \mapsto$$

$$\underline{\varepsilon}_m = \underline{E} \underline{m}$$

VETTORE
DELLA
DEFORMAZIONE

FORMULA DI
CAUCHY PER LA
DEFORMAZIONE
(qualunque δ)



CALCOLO DELLE COMPONENTI

$$\underline{\varepsilon}_m = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ 1/2 \gamma_{xy} & \varepsilon_y & 1/2 \gamma_{yz} \\ 1/2 \gamma_{xz} & 1/2 \gamma_{yz} & \varepsilon_z \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \alpha \varepsilon_x + 1/2 \gamma_{xy} \beta + 1/2 \gamma_{xz} \gamma \\ 1/2 \gamma_{xy} \alpha + \beta \varepsilon_y + 1/2 \gamma_{yz} \gamma \\ 1/2 \gamma_{xz} \alpha + 1/2 \gamma_{yz} \beta + \gamma \varepsilon_z \end{bmatrix} \equiv \begin{bmatrix} \varepsilon_{mx} \\ \varepsilon_{my} \\ \varepsilon_{mz} \end{bmatrix}$$

FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\underline{\underline{PQ}}$: FIBRA GENERICA $\equiv \underline{\underline{m}}_\delta = \delta \underline{\underline{m}} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{\underline{u}}_Q = \underbrace{\underline{\underline{u}}_P}_{\textcircled{1} \text{ RIGIDA}} + \underbrace{\underline{\underline{\Omega}} d\underline{\underline{x}}}_{\textcircled{2} \text{ RIGIDA}} + \underbrace{\underline{\underline{E}} d\underline{\underline{x}}}_{\textcircled{3} \text{ AFFINE}}$$

SPOSTAMENTO DI Q:

$$\underline{\underline{u}}_Q = \underline{\underline{u}}_P + \underline{\underline{E}} \underline{\underline{m}}_\delta + \underline{\underline{\Omega}} \underline{\underline{m}}_\delta = \underline{\underline{u}}_P + \delta \underline{\underline{E}} \underline{\underline{m}} + \cancel{\underline{\underline{\Omega}} \underline{\underline{m}}_\delta}$$

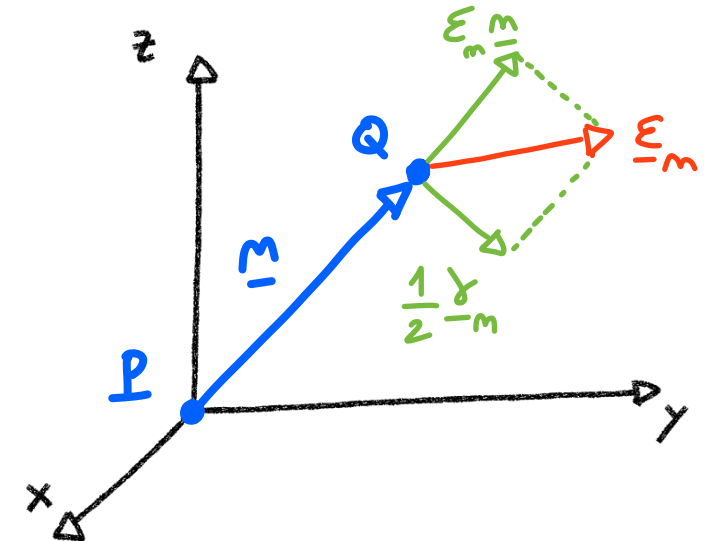
DEFORMAZIONE
PURA

$$\delta \underline{\underline{E}} \underline{\underline{m}} \mapsto$$

$$\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}}$$

VETTORE
DELLA
DEFORMAZIONE

FORMULA DI
CAUCHY PER LA
DEFORMAZIONE
(qualunque δ)



CALCOLO DELLE COMPONENTI

$$\underline{\underline{\varepsilon}}_m = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ 1/2 \gamma_{xy} & \varepsilon_y & 1/2 \gamma_{yz} \\ 1/2 \gamma_{xz} & 1/2 \gamma_{yz} & \varepsilon_z \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \alpha \varepsilon_x + 1/2 \gamma_{xy} \beta + 1/2 \gamma_{xz} \gamma \\ 1/2 \gamma_{xy} \alpha + \beta \varepsilon_y + 1/2 \gamma_{yz} \gamma \\ 1/2 \gamma_{xz} \alpha + 1/2 \gamma_{yz} \beta + \gamma \varepsilon_z \end{bmatrix} \equiv \begin{bmatrix} \varepsilon_{mx} \\ \varepsilon_{my} \\ \varepsilon_{mz} \end{bmatrix}$$

DECOMPOSIZIONE IN DILATAZIONI E SCORRIMENTI:

$$\underline{\underline{m}} = \underline{\underline{\varepsilon}}_m \underline{\underline{m}} + \frac{1}{2} \underline{\underline{\gamma}}_m$$

- PRIMA SI CALCOLA $\underline{\underline{\varepsilon}}_m$
- POI PER DIFFERENZA $\underline{\underline{\gamma}}_m$

FORMULA DI CAUCHY PER LA DEFORMAZIONE

- $\underline{\underline{PQ}}$: FIBRA GENERICA $\equiv \underline{\underline{m}}_\delta = \delta \underline{\underline{m}} = \delta [\alpha \ \beta \ \gamma]^T$
 - $\delta \ll 1$; $\alpha^2 + \beta^2 + \gamma^2 = 1$
- $$\underline{\underline{u}}_Q = \underbrace{\underline{\underline{u}}_P}_{\text{① RIGIDA}} + \underbrace{\underline{\underline{\Omega}} d\underline{\underline{x}}}_{\text{② RIGIDA}} + \underbrace{\underline{\underline{E}} d\underline{\underline{x}}}_{\text{③ AFFINE}}$$

SPOSTAMENTO DI Q:

$$\underline{\underline{u}}_Q = \underline{\underline{u}}_P + \underline{\underline{E}} \underline{\underline{m}}_\delta + \underline{\underline{\Omega}} \underline{\underline{m}}_\delta = \underline{\underline{u}}_P + \delta \underline{\underline{E}} \underline{\underline{m}} + \cancel{\delta \underline{\underline{\Omega}} \underline{\underline{m}}}$$

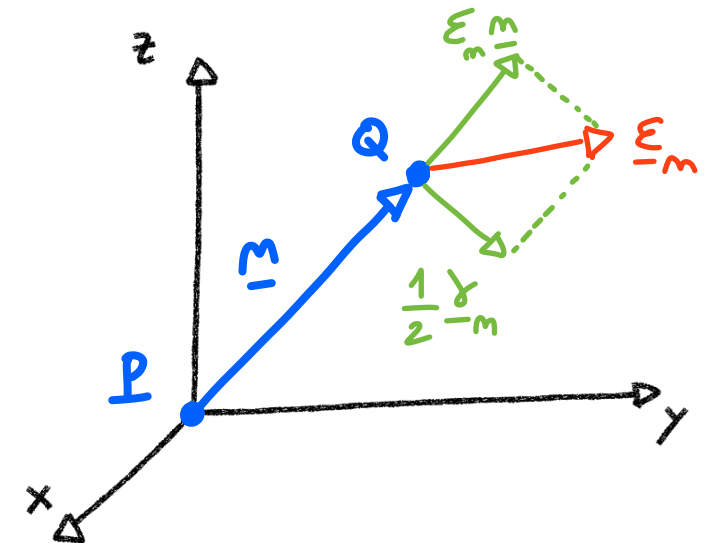
DEFORMAZIONE
PURA

$$\delta \underline{\underline{E}} \underline{\underline{m}} \mapsto$$

$$\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}}$$

VETTORE
DELLA
DEFORMAZIONE

FORMULA DI
CAUCHY PER LA
DEFORMAZIONE
(qualunque δ)



CALCOLO DELLE COMPONENTI

$$\underline{\underline{\varepsilon}}_m = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ 1/2 \gamma_{xy} & \varepsilon_y & 1/2 \gamma_{yz} \\ 1/2 \gamma_{xz} & 1/2 \gamma_{yz} & \varepsilon_z \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \alpha \varepsilon_x + 1/2 \gamma_{xy} \beta + 1/2 \gamma_{xz} \gamma \\ 1/2 \gamma_{xy} \alpha + \beta \varepsilon_y + 1/2 \gamma_{yz} \gamma \\ 1/2 \gamma_{xz} \alpha + 1/2 \gamma_{yz} \beta + \gamma \varepsilon_z \end{bmatrix} \equiv \begin{bmatrix} \varepsilon_{mx} \\ \varepsilon_{my} \\ \varepsilon_{mz} \end{bmatrix}$$

DECOMPOSIZIONE IN DILATAZIONI E SCORRIMENTI:

$$\underline{\underline{m}} = \underline{\underline{\varepsilon}}_m \underline{\underline{m}} + \frac{1}{2} \underline{\underline{\gamma}}_m$$

- PRIMA SI CALCOLA $\underline{\underline{\varepsilon}}_m$ $\Rightarrow$
- POI PER DIFFERENZA $\underline{\underline{\gamma}}_m$

PROBLEMA
AGLI

AUTOVALORI

PROBLEMA AGLI AUTOVALORI

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

H_p. : $\varepsilon_m = E_m = \left(\frac{m}{3} \right)^2$

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} = (\underline{\underline{\varepsilon}}_m) \underline{\underline{m}}$

$$\underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{m}} = \underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{I}} \underline{\underline{m}} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$
$$\rightarrow (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} = (\varepsilon_3) \underline{\underline{m}}$

$$\underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{m}} = \underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{I}} \underline{\underline{m}} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$\rightarrow (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$[E] = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y & 1/2 \gamma_{yz} \\ \text{sym} & & \varepsilon_z \end{bmatrix}$$

$$[I] = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ \text{sym} & & 1 \end{bmatrix}$$

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\varepsilon}_m = \underline{\underline{E}} \underline{m} = \underline{\varepsilon_m} \underline{m}$

$$\underline{\underline{E}} \underline{m} - \varepsilon_m \underline{m} = \underline{\underline{E}} \underline{m} - \varepsilon_m \underline{\underline{I}} \underline{m} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{m} = \underline{0}$$

$$\rightarrow (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{m} = \underline{0}$$

$$\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}} = \begin{bmatrix} \varepsilon_x - \varepsilon_m & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y - \varepsilon_m & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z - \varepsilon_m \end{bmatrix}$$

$$[E] = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$

$$[I] = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ \text{SYM} & & 1 \end{bmatrix}$$

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} = (\varepsilon_m) \underline{\underline{m}}$

$$\underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{m}} = \underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{I}} \underline{\underline{m}} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$\Rightarrow (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}} = \begin{bmatrix} \varepsilon_x - \varepsilon_m & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y - \varepsilon_m & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z - \varepsilon_m \end{bmatrix}$$

$$[E] = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$

$$[I] = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ \text{SYM} & & 1 \end{bmatrix}$$

SOLUZIONE
NON
BANALE :

$$\det(\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) = 0$$

EQUAZIONE
CARATTERISTICA
SECOLARE (eq. scalare 3°)

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} = (\varepsilon_m) \underline{\underline{m}}$

$$\underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{m}} = \underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{I}} \underline{\underline{m}} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}} = \begin{bmatrix} \varepsilon_x - \varepsilon_m & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y - \varepsilon_m & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z - \varepsilon_m \end{bmatrix}$$

$$[E] = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$

$$[I] = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ \text{SYM} & & 1 \end{bmatrix}$$

SOLUZIONE
NON
BANALE

$\det(\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) = 0$ $\left\{ \begin{array}{l} \text{EQUAZIONE} \\ \text{CARATTERISTICA} \\ \text{SECOLARE} \end{array} \right.$ (eq. scalare 3°)

$$\varepsilon_m^3 - \bar{I}_1 \varepsilon_m + \bar{I}_2 \varepsilon_m - \bar{I}_3 = 0$$

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} = \underline{\underline{\varepsilon}}_m \underline{\underline{m}}$

$$\underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{m}} = \underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{I}} \underline{\underline{m}} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}} = \begin{bmatrix} \varepsilon_x - \varepsilon_m & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y - \varepsilon_m & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z - \varepsilon_m \end{bmatrix}$$

$$[\underline{\underline{E}}] = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$

$$[\underline{\underline{I}}] = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ \text{SYM} & & 1 \end{bmatrix}$$

SOLUZIONE
NON
BANALE

$\det(\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) = 0$ $\left\{ \begin{array}{l} \text{EQUAZIONE} \\ \text{CARATTERISTICA} \\ \text{SECOLARE} \end{array} \right.$ (eq. scalare 3°)

$$\varepsilon_m^3 - \bar{I}_1 \varepsilon_m^2 + \bar{I}_2 \varepsilon_m - \bar{I}_3 = 0$$

INVARIANTI
DI

DEFORMAZIONE

$$\bar{I}_1 = \text{tr}(\underline{\underline{E}}) = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

$$\bar{I}_2 = \frac{1}{2} (\text{tr}^2(\underline{\underline{E}}) - \text{tr}(\underline{\underline{E}}^2)) = \varepsilon_x \varepsilon_y + \varepsilon_x \varepsilon_z + \varepsilon_y \varepsilon_z - \frac{1}{4} (\gamma_{xy}^2 + \gamma_{xz}^2 + \gamma_{yz}^2)$$

$$\bar{I}_3 = \det(\underline{\underline{E}})$$

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} = (\underline{\underline{\varepsilon}}_m) \underline{\underline{m}} \rightarrow (\underline{\underline{E}} - \underline{\underline{\varepsilon}}_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$

$$\underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{m}} = \underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{I}} \underline{\underline{m}} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}} = \begin{bmatrix} \varepsilon_x - \varepsilon_m & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y - \varepsilon_m & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z - \varepsilon_m \end{bmatrix}$$

$$[\underline{\underline{E}}] = \begin{bmatrix} \varepsilon_x & 1/2 \gamma_{xy} & 1/2 \gamma_{xz} \\ & \varepsilon_y & 1/2 \gamma_{yz} \\ \text{SYM} & & \varepsilon_z \end{bmatrix}$$

$$[\underline{\underline{I}}] = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ \text{SYM} & & 1 \end{bmatrix}$$

SOLUZIONE
NON
BANALE

$\det(\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) = 0$ $\left\{ \begin{array}{l} \text{EQUAZIONE} \\ \text{CARATTERISTICA} \\ \text{SECOLARE} \end{array} \right.$ (eq. scalare 3°)

$$\varepsilon_m^3 - \bar{I}_1 \varepsilon_m + \bar{I}_2 \varepsilon_m - \bar{I}_3 = 0$$

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a(ei - hf) + b(di - fg) + c(dh - eg)$$

INVARIANTI
DI
DEFORMAZIONE

$$\bar{I}_1 = \text{tr}(\underline{\underline{E}}) = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

$$\bar{I}_2 = \frac{1}{2} (\text{tr}^2(\underline{\underline{E}}) - \text{tr}(\underline{\underline{E}}^2)) = \varepsilon_x \varepsilon_y + \varepsilon_x \varepsilon_z + \varepsilon_y \varepsilon_z - \frac{1}{4} (\gamma_{xy}^2 + \gamma_{xz}^2 + \gamma_{yz}^2)$$

$$\bar{I}_3 = \det(\underline{\underline{E}})$$

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} \equiv \varepsilon_m \underline{\underline{m}} \quad \mapsto \quad (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$

$$\underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{m}} = \underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{I}} \underline{\underline{m}} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$\det(\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) = 0 \quad \left\{ \begin{array}{l} \text{EQUAZIONE} \\ \text{CARATTERISTICA} \\ \text{SCOLARE} \end{array} \right. \quad (\text{eq. scalare } 3^{\circ})$$

$$\varepsilon_m^3 - \bar{I}_1 \varepsilon_m + \bar{I}_2 \varepsilon_m - \bar{I}_3 = 0$$

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a(ei - hf) + b(di - fg) + c(dh - eg)$$

INVARIANTI
DI
DEFORMAZIONE

$$\left\{ \begin{array}{l} \bar{I}_1 = \text{tr}(\underline{\underline{E}}) = \varepsilon_x + \varepsilon_y + \varepsilon_z \\ \bar{I}_2 = \frac{1}{2}(\text{tr}^2(\underline{\underline{E}}) - \text{tr}(\underline{\underline{E}}^2)) = \varepsilon_x \varepsilon_y + \varepsilon_x \varepsilon_z + \varepsilon_y \varepsilon_z - \frac{1}{4}(\gamma_{xy}^2 + \gamma_{xz}^2 + \gamma_{yz}^2) \\ \bar{I}_3 = \det(\underline{\underline{E}}) \end{array} \right.$$

PROBLEMA AGLI AUTOVALORI

CALCOLO ε_m :

Hip.: $\underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} \equiv \varepsilon_m \underline{\underline{m}} \quad \rightarrow \quad (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$

$$\underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{m}} = \underline{\underline{E}} \underline{\underline{m}} - \varepsilon_m \underline{\underline{I}} \underline{\underline{m}} = (\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) \underline{\underline{m}} = \underline{\underline{0}}$$

$$\det(\underline{\underline{E}} - \varepsilon_m \underline{\underline{I}}) = 0 \quad \left\{ \begin{array}{l} \text{EQUAZIONE} \\ \text{CARATTERISTICA} \\ \text{SCOLARE} \end{array} \right. \quad (\text{eq. scalare } 3^{\circ})$$

$$\varepsilon_m^3 - \bar{I}_1 \varepsilon_m + \bar{I}_2 \varepsilon_m - \bar{I}_3 = 0$$

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a(ei - hf) + b(di - fg) + c(dh - eg)$$

INVARIANTI
DI
DEFORMAZIONE

$$\left\{ \begin{array}{l} \bar{I}_1 = \text{tr}(\underline{\underline{E}}) = \varepsilon_x + \varepsilon_y + \varepsilon_z \\ \bar{I}_2 = \frac{1}{2}(\text{tr}^2(\underline{\underline{E}}) - \text{tr}(\underline{\underline{E}}^2)) = \varepsilon_x \varepsilon_y + \varepsilon_x \varepsilon_z + \varepsilon_y \varepsilon_z - \frac{1}{4}(\gamma_{xy}^2 + \gamma_{xz}^2 + \gamma_{yz}^2) \\ \bar{I}_3 = \det(\underline{\underline{E}}) \end{array} \right.$$

DEFORMAZIONI
PRINCIPALI

$$\varepsilon_1, \varepsilon_2, \varepsilon_3$$

RADICI REALI
DELL' eq. CARATT.
AUTOVALORI

PROBLEMA AGLI AUTOVETTORI

■

PROBLEMA AGLI AUTOVETTORI

CALCOLO DIREZIONI
PRINCIPALI DI DEFORM.

:

PROBLEMA AGLI
AUTOVETTORI

■

PROBLEMA AGLI AUTOVETTORI

CALCOLO DIREZIONI
PRINCIPALI DI DEFORM. :

PROBLEMA AGLI
AUTOVETTORI

- NOTO ϵ_i , $i = 1, 2, 3$
(DEFORMAZIONE PRINCIPALE)

■

PROBLEMA AGLI AUTOVETTORI

CALCOLO DIREZIONI
PRINCIPALI DI DEFORM.

PROBLEMA AGLI
AUTOVETTORI

- NOTO ε_i , $i = 1, 2, 3$
(DEFORMAZIONE PRINCIPALE)

- IMPONGO $\forall \varepsilon_i$ CHE $\underline{m}_i = [\alpha_i \ \beta_i \ \gamma_i]^T$ E CHE :

$$\begin{cases} \underline{\underline{E}} \underline{m}_i = \varepsilon_i \underline{m}_i \\ \underline{m}_i \cdot \underline{m}_i^T = 1 \end{cases} \rightarrow \begin{cases} \underline{\underline{E}} \underline{m}_i = \varepsilon_i \underline{m}_i \\ \alpha_i^2 + \beta_i^2 + \gamma_i^2 = 1 \end{cases}$$

PROBLEMA AGLI AUTOVETTORI

CALCOLO DIREZIONI
PRINCIPALI DI DEFORM.

PROBLEMA AGLI
AUTOVETTORI

- NOTO ϵ_i , $i = 1, 2, 3$
(DEFORMAZIONE PRINCIPALE)

- IMPONGO $\forall \epsilon_i$ CHE $\underline{m}_i = [\alpha_i \ \beta_i \ \gamma_i]^T$ E CHE :

$$\begin{cases} \underline{\underline{E}} \underline{m}_i = \epsilon_i \underline{m}_i \\ \underline{m}_i \cdot \underline{m}_i^T = 1 \end{cases} \rightarrow \begin{cases} \underline{\underline{E}} \underline{m}_i = \epsilon_i \underline{m}_i \\ \alpha_i^2 + \beta_i^2 + \gamma_i^2 = 1 \end{cases}$$

Condizione di "DIAGONALITÀ"

Condizione di "ORTONORMALITÀ"

PROBLEMA AGLI AUTOVETTORI

CALCOLO DIREZIONI
PRINCIPALI DI DEFORM.

PROBLEMA AGLI
AUTOVETTORI

- NOTO ϵ_i , $i = 1, 2, 3$
(DEFORMAZIONE PRINCIPALE)

- IMPONGO $\forall \epsilon_i$ CHE $\underline{m}_i = [\alpha_i \beta_i \gamma_i]^T$ E CHE:

$$\begin{cases} \underline{\underline{E}} \underline{m}_i = \epsilon_i \underline{m}_i \\ \underline{m}_i \cdot \underline{m}_i^T = 1 \end{cases} \rightarrow \begin{cases} \underline{\underline{E}} \underline{m}_i = \epsilon_i \underline{m}_i \\ \alpha_i^2 + \beta_i^2 + \gamma_i^2 = 1 \end{cases}$$

Condizione di "DIAGONALITÀ"

Condizione di "ORTONORMALITÀ"

BASE DI
RIFERIMENTO
"PRINCIPALE"

PROBLEMA AGLI AUTOVETTORI

CALCOLO DIREZIONI
PRINCIPALI DI DEFORM.

PROBLEMA AGLI
AUTOVETTORI

- NOTO ϵ_i , $i = 1, 2, 3$
(DEFORMAZIONE PRINCIPALE)

- IMPONGO $\forall \epsilon_i$ CHE $\underline{m}_i = [\alpha_i \beta_i \gamma_i]^T$ E CHE:

$$\begin{cases} \underline{\underline{E}} \underline{m}_i = \epsilon_i \underline{m}_i \\ \underline{m}_i \cdot \underline{m}_i^T = 1 \end{cases} \rightarrow \begin{cases} \underline{\underline{E}} \underline{m}_i = \epsilon_i \underline{m}_i \\ \alpha_i^2 + \beta_i^2 + \gamma_i^2 = 1 \end{cases}$$

Condizione di "DIAGONALITÀ"

Condizione di "ORTONORMALITÀ"

BASE DI
RIFERIMENTO
"PRINCIPALE"

AUTOVETTORI ASSOCIATI
AGLI AUTOVALORI

$(\epsilon_i, \underline{m}_i) \Leftrightarrow \underline{\underline{E}} \underline{m}_i$ SONO $\perp$

CLASSIFICAZIONE STATO DEFORMATIVO

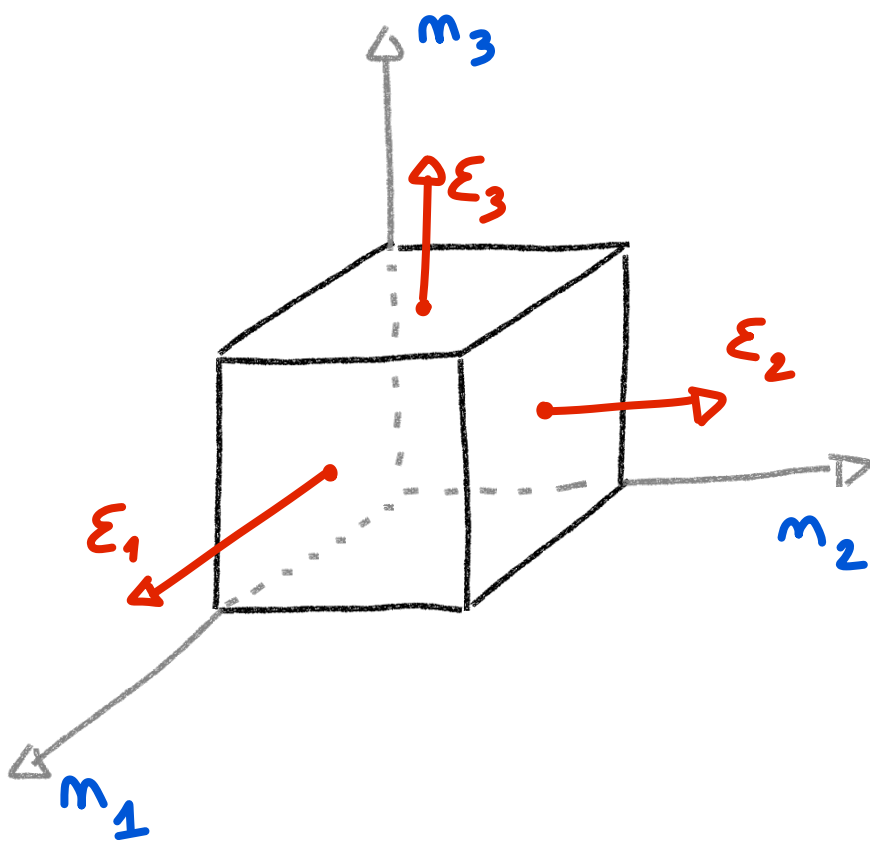
La classificazione si basa sulla MOLTEPLICITÀ degli AUTOVALORI ϵ_i

CLASSIFICAZIONE STATO DEFORMATIVO

La classificazione si basa sulla MOLTEPLICITÀ degli AUTOVALORI ϵ_i

TRIASSIALE :
(CUBICO)

$$\epsilon_1 \neq \epsilon_2 \neq \epsilon_3$$



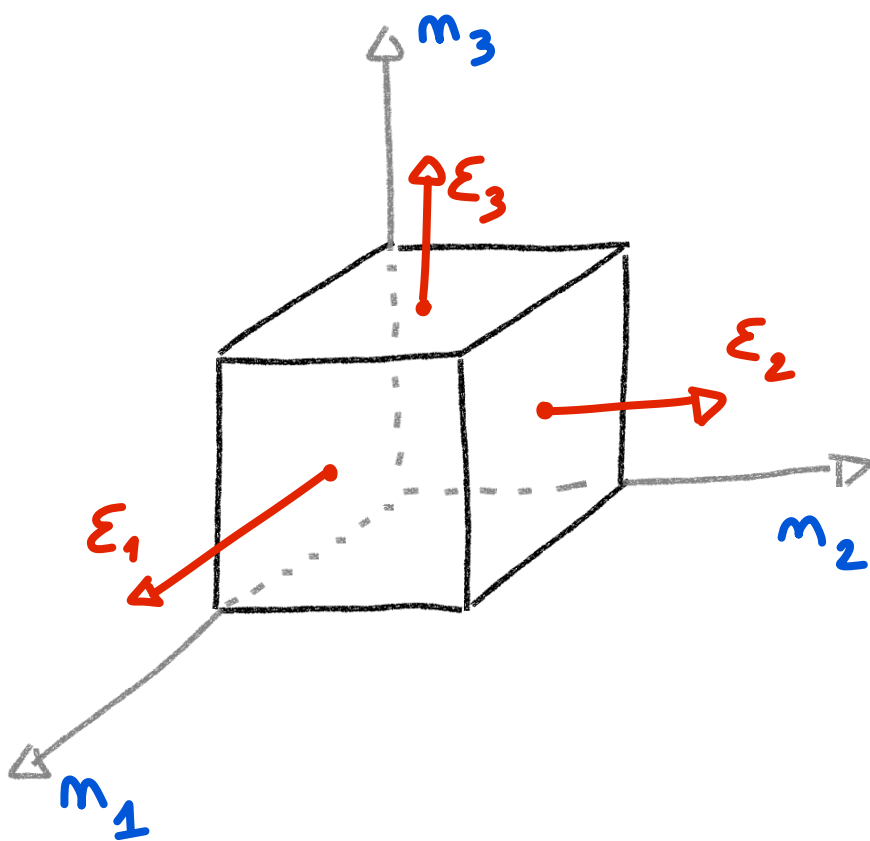
- STESSA FORMA
- VOLUME DIVERSO

CLASSIFICAZIONE STATO DEFORMATIVO

La classificazione si basa sulla MOLTEPLICITÀ degli AUTOVALORI ϵ_i

TRIASSIALE :
(CUBICO)

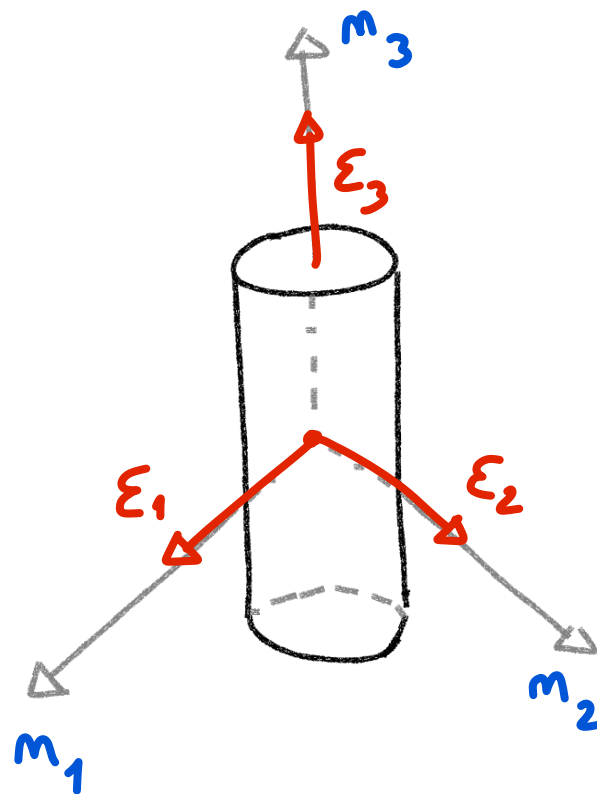
$$\epsilon_1 \neq \epsilon_2 \neq \epsilon_3$$



- STESSA FORMA
- VOLUME DIVERSO

CILINDRICO :

$$\epsilon_1 = \epsilon_2 \neq \epsilon_3$$



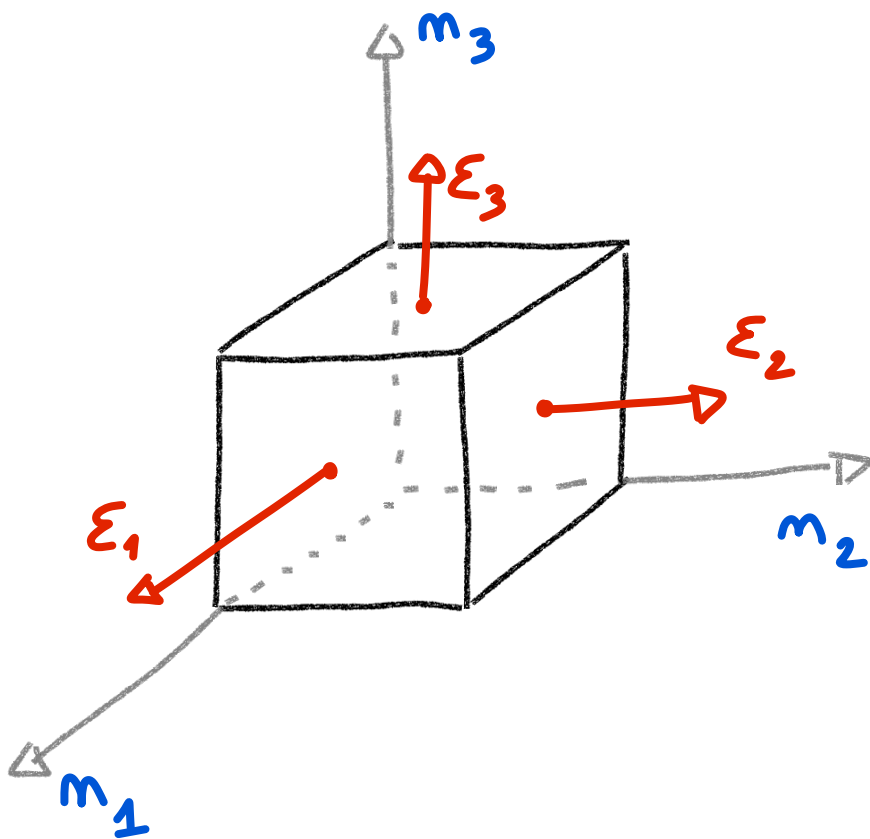
- INFINITE DIREZIONI

CLASSIFICAZIONE STATO DEFORMATIVO

La classificazione si basa sulla MOLTEPLICITÀ degli AUTOVALORI ϵ_i :

TRIASSIALE :
(CUBICO)

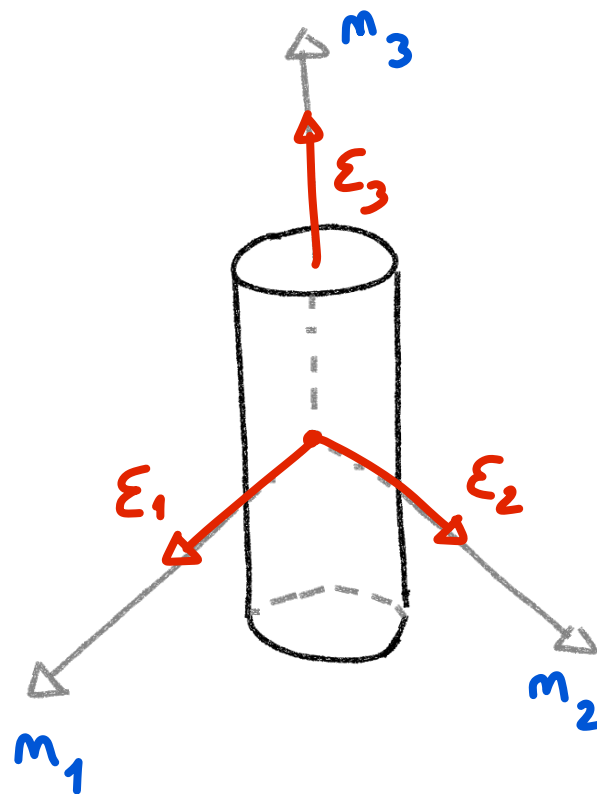
$$\epsilon_1 \neq \epsilon_2 \neq \epsilon_3$$



- STESSA FORMA
- VOLUME DIVERSO

CILINDRICO :

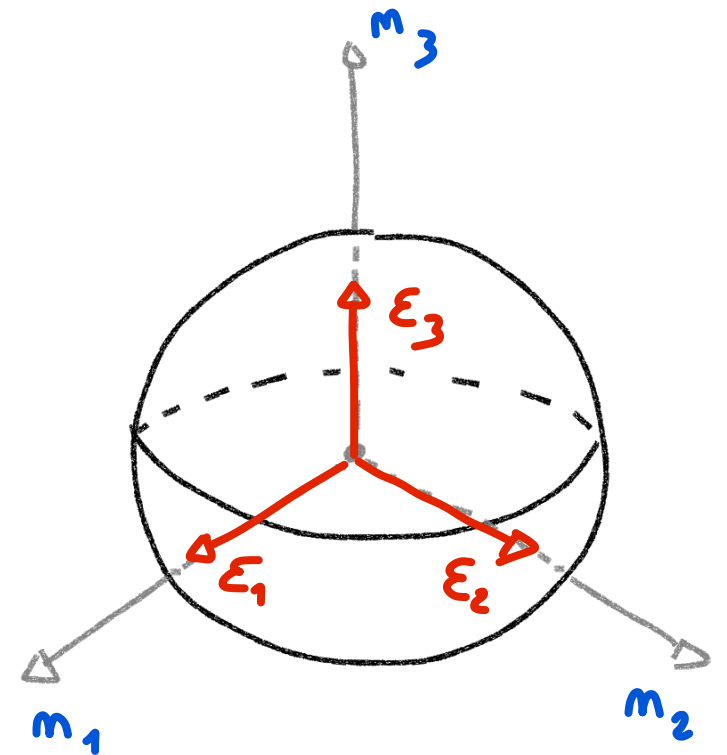
$$\epsilon_1 = \epsilon_2 \neq \epsilon_3$$



- INFINITE DIREZIONI

SFERICO :
(IDROSTATICO)

$$\epsilon_1 = \epsilon_2 = \epsilon_3$$



CLASSIFICAZIONE STATO DEFORMATIVO

RIFERIMENTO PRINCIPALE

■

CLASSIFICAZIONE STATO DEFORMATIVO

RIFERIMENTO PRINCIPALE

TERNA
PRINCIPALE
DI
DEFORMAZIONE

$$: x_1, x_2, x_3 \rightarrow \underline{\underline{E}} = \frac{1}{2} (\underline{\underline{F}} + \underline{\underline{F}}^T) = \begin{bmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix}$$

TENSORE
DELLA
DEFORMAZIONE
DIAGONALE

CLASSIFICAZIONE STATO DEFORMATIVO

RIFERIMENTO PRINCIPALE

TERNA
PRINCIPALE
DI
DEFORMAZIONE

$$: x_1, x_2, x_3 \rightarrow \underline{\underline{E}} = \frac{1}{2}(\underline{\underline{F}} + \underline{\underline{F}}^T) = \begin{bmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix}$$

TENSORE
DELLA
DEFORMAZIONE
DIAGONALE

↳

$$\underline{\underline{\varepsilon}}_m = \underline{\underline{E}}_m = \begin{bmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \alpha \varepsilon_1 \\ \beta \varepsilon_2 \\ \gamma \varepsilon_3 \end{bmatrix} = \varepsilon_m \underline{\underline{m}} + \frac{1}{2} \underline{\underline{\gamma}}_m$$

CLASSIFICAZIONE STATO DEFORMATIVO

RIFERIMENTO PRINCIPALE

TERNA
PRINCIPALE
DI
DEFORMAZIONE

$$: x_1, x_2, x_3 \rightarrow \underline{\underline{E}} = \frac{1}{2}(\underline{\underline{F}} + \underline{\underline{F}}^T) = \begin{bmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix}$$

TENSORE
DELLA
DEFORMAZIONE
DIAGONALE

$$\downarrow \quad \underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} = \begin{bmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \alpha \varepsilon_1 \\ \beta \varepsilon_2 \\ \gamma \varepsilon_3 \end{bmatrix} = \varepsilon_m \underline{\underline{m}} + \frac{1}{2} \underline{\underline{\gamma}}_m$$

(*) DATO CHE LA PROIEZIONE SU $\underline{\underline{m}}$ RESTITUISCE SOLO ε_m E NON $\underline{\underline{\gamma}}_m$!

$$\varepsilon_m = \underline{\underline{m}}^T \cdot \underline{\underline{\varepsilon}}_m = [\alpha \ \beta \ \gamma] \begin{bmatrix} \alpha \varepsilon_1 \\ \beta \varepsilon_2 \\ \gamma \varepsilon_3 \end{bmatrix} = \underline{\underline{\alpha^2 \varepsilon_1 + \beta^2 \varepsilon_2 + \gamma^2 \varepsilon_3}}$$

CLASSIFICAZIONE STATO DEFORMATIVO

RIFERIMENTO PRINCIPALE

TERNA
PRINCIPALE
DI
DEFORMAZIONE

$$: x_1, x_2, x_3 \rightarrow \underline{\underline{E}} = \frac{1}{2}(\underline{\underline{F}} + \underline{\underline{F}}^T) = \begin{bmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix}$$

TENSORE
DELLA
DEFORMAZIONE
DIAGONALE

$$\downarrow \quad \underline{\underline{\varepsilon}}_m = \underline{\underline{E}} \underline{\underline{m}} = \begin{bmatrix} \varepsilon_1 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_3 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \alpha \varepsilon_1 \\ \beta \varepsilon_2 \\ \gamma \varepsilon_3 \end{bmatrix} = \varepsilon_m \underline{\underline{m}} + \frac{1}{2} \underline{\underline{\gamma}}_m$$

(*) DATO CHE LA PROIEZIONE SU $\underline{\underline{m}}$ RESTITUISCE SOLO ε_m E NON $\underline{\underline{\gamma}}_m$!

$$\varepsilon_m = \underline{\underline{m}}^T \cdot \underline{\underline{\varepsilon}}_m = [\alpha \ \beta \ \gamma] \begin{bmatrix} \alpha \varepsilon_1 \\ \beta \varepsilon_2 \\ \gamma \varepsilon_3 \end{bmatrix} = \underline{\underline{\alpha^2 \varepsilon_1 + \beta^2 \varepsilon_2 + \gamma^2 \varepsilon_3}}$$

(*) $\underline{\underline{\gamma}}_m$ SI RICAVA PER DIFFERENZA:

$$\underline{\underline{\gamma}}_m = 2 \left(\underline{\underline{\varepsilon}}_m - \varepsilon_m \underline{\underline{m}} \right)$$

CLASSIFICAZIONE STATO DEFORMATIVO

CIRCONFERENZE DI MOHR PER LA DEFORMAZIONE

CLASSIFICAZIONE STATO DEFORMATIVO

CIRCONFERENZE DI MOHR PER LA DEFORMAZIONE

$$\left\{ \begin{array}{l} \varepsilon_m = \alpha^2 \varepsilon_1 + \beta^2 \varepsilon_2 + \gamma^2 \varepsilon_3 \\ |\underline{\varepsilon}_m| = \varepsilon_m^2 + \frac{\gamma_m^2}{4} = \varepsilon_1^2 \gamma^2 + \varepsilon_2^2 \beta^2 + \varepsilon_3^2 \alpha^2 \\ \alpha^2 + \beta^2 + \gamma^2 = 1 \end{array} \right.$$

CLASSIFICAZIONE STATO DEFORMATIVO

CIRCONFERENZE DI MOHR PER LA DEFORMAZIONE

$$\begin{cases} \varepsilon_m = \alpha^2 \varepsilon_1 + \beta^2 \varepsilon_2 + \gamma^2 \varepsilon_3 \\ |\underline{\varepsilon}_m| = \varepsilon_m^2 + \frac{\gamma_m^2}{4} = \varepsilon_1^2 \gamma^2 + \varepsilon_2^2 \beta^2 + \varepsilon_3^2 \gamma^2 \\ \alpha^2 + \beta^2 + \gamma^2 = 1 \end{cases} \Rightarrow \begin{cases} \alpha^2 = \bar{A}_1(\varepsilon_m, \gamma_m)/a_1 \\ \beta^2 = \bar{A}_2(\varepsilon_m, \gamma_m)/a_2 \\ \gamma^2 = \bar{A}_3(\varepsilon_m, \gamma_m)/a_3 \end{cases}$$

$$\alpha^2 = \frac{(\varepsilon_m - \varepsilon_2)(\varepsilon_m - \varepsilon_3) + \frac{\gamma_m^2}{4}}{(\varepsilon_1 - \varepsilon_2)(\varepsilon_1 - \varepsilon_3)}$$

Soluzioni rispetto a
 α, β, γ

$$\beta^2 = \frac{(\varepsilon_m - \varepsilon_1)(\varepsilon_m - \varepsilon_3) + \frac{\gamma_m^2}{4}}{(\varepsilon_2 - \varepsilon_1)(\varepsilon_2 - \varepsilon_3)}$$

$$\gamma^2 = \frac{(\varepsilon_m - \varepsilon_1)(\varepsilon_m - \varepsilon_2) + \frac{\gamma_m^2}{4}}{(\varepsilon_3 - \varepsilon_1)(\varepsilon_3 - \varepsilon_2)}$$

CLASSIFICAZIONE STATO DEFORMATIVO

CIRCONFERENZE DI MOHR PER LA DEFORMAZIONE

$$\begin{cases} \varepsilon_m = \alpha^2 \varepsilon_1 + \beta^2 \varepsilon_2 + \gamma^2 \varepsilon_3 \\ |\underline{\varepsilon}_m| = \varepsilon_m^2 + \frac{\gamma_m^2}{4} = \varepsilon_1^2 \gamma^2 + \varepsilon_2^2 \beta^2 + \varepsilon_3^2 \gamma^2 \\ \alpha^2 + \beta^2 + \gamma^2 = 1 \end{cases} \Rightarrow \begin{cases} \alpha^2 = \bar{A}_1(\varepsilon_m, \gamma_m) / a_1 \\ \beta^2 = \bar{A}_2(\varepsilon_m, \gamma_m) / a_2 \\ \gamma^2 = \bar{A}_3(\varepsilon_m, \gamma_m) / a_3 \end{cases}$$

$$H_p: \varepsilon_1 > \varepsilon_2 > \varepsilon_3$$

$$a_1 = (\varepsilon_1 - \varepsilon_2)(\varepsilon_1 - \varepsilon_3) > 0 \Rightarrow \bar{A}_1(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_2)(\varepsilon_m - \varepsilon_3) + \gamma_m^2 / 4 \geq 0$$

$$a_2 = (\varepsilon_2 - \varepsilon_1)(\varepsilon_2 - \varepsilon_3) < 0 \Rightarrow \bar{A}_2(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_1)(\varepsilon_m - \varepsilon_3) + \gamma_m^2 / 4 \leq 0$$

$$a_3 = (\varepsilon_3 - \varepsilon_1)(\varepsilon_3 - \varepsilon_2) > 0 \Rightarrow \bar{A}_3(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_1)(\varepsilon_m - \varepsilon_2) + \gamma_m^2 / 4 \geq 0$$

CLASSIFICAZIONE STATO DEFORMATIVO

CIRCONFERENZE DI MOHR PER LA DEFORMAZIONE

$$\begin{cases} \varepsilon_m = \alpha^2 \varepsilon_1 + \beta^2 \varepsilon_2 + \gamma^2 \varepsilon_3 \\ |\underline{\varepsilon}_m| = \varepsilon_m^2 + \frac{\gamma_m^2}{4} = \varepsilon_1^2 \gamma^2 + \varepsilon_2^2 \beta^2 + \varepsilon_3^2 \gamma^2 \\ \alpha^2 + \beta^2 + \gamma^2 = 1 \end{cases} \Rightarrow \begin{cases} \alpha^2 = \bar{A}_1(\varepsilon_m, \gamma_m)/a_1 \\ \beta^2 = \bar{A}_2(\varepsilon_m, \gamma_m)/a_2 \\ \gamma^2 = \bar{A}_3(\varepsilon_m, \gamma_m)/a_3 \end{cases}$$

$$H_p: \varepsilon_1 > \varepsilon_2 > \varepsilon_3$$

$$a_1 = (\varepsilon_1 - \varepsilon_2)(\varepsilon_1 - \varepsilon_3) > 0 \Rightarrow \bar{A}_1(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_2)(\varepsilon_m - \varepsilon_3) + \gamma_m^2/4 \geq 0$$

$$a_2 = (\varepsilon_2 - \varepsilon_1)(\varepsilon_2 - \varepsilon_3) < 0 \Rightarrow \bar{A}_2(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_1)(\varepsilon_m - \varepsilon_3) + \gamma_m^2/4 \leq 0$$

$$a_3 = (\varepsilon_3 - \varepsilon_1)(\varepsilon_3 - \varepsilon_2) > 0 \Rightarrow \bar{A}_3(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_1)(\varepsilon_m - \varepsilon_2) + \gamma_m^2/4 \geq 0$$

CIRCONFERENZE
DI
MOHR : $\bar{C}_1, \bar{C}_2, \bar{C}_3 = 0$

CLASSIFICAZIONE STATO DEFORMATIVO

CIRCONFERENZE DI MOHR PER LA DEFORMAZIONE

$$\begin{cases} \varepsilon_m = \alpha^2 \varepsilon_1 + \beta^2 \varepsilon_2 + \gamma^2 \varepsilon_3 \\ |\underline{\varepsilon}_m| = \varepsilon_m^2 + \frac{\gamma_m^2}{4} = \varepsilon_1^2 \gamma^2 + \varepsilon_2^2 \beta^2 + \varepsilon_3^2 \gamma^2 \\ \alpha^2 + \beta^2 + \gamma^2 = 1 \end{cases} \Rightarrow \begin{cases} \alpha^2 = \bar{A}_1(\varepsilon_m, \gamma_m) / a_1 \\ \beta^2 = \bar{A}_2(\varepsilon_m, \gamma_m) / a_2 \\ \gamma^2 = \bar{A}_3(\varepsilon_m, \gamma_m) / a_3 \end{cases}$$

$$H_p: \varepsilon_1 > \varepsilon_2 > \varepsilon_3$$

$$a_1 = (\varepsilon_1 - \varepsilon_2)(\varepsilon_1 - \varepsilon_3) > 0 \Rightarrow \bar{A}_1(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_2)(\varepsilon_m - \varepsilon_3) + \gamma_m^2 / 4 \geq 0$$

$$a_2 = (\varepsilon_2 - \varepsilon_1)(\varepsilon_2 - \varepsilon_3) < 0 \Rightarrow \bar{A}_2(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_1)(\varepsilon_m - \varepsilon_3) + \gamma_m^2 / 4 \leq 0$$

$$a_3 = (\varepsilon_3 - \varepsilon_1)(\varepsilon_3 - \varepsilon_2) > 0 \Rightarrow \bar{A}_3(\varepsilon_m, \gamma_m) = (\varepsilon_m - \varepsilon_1)(\varepsilon_m - \varepsilon_2) + \gamma_m^2 / 4 \geq 0$$

CIRCONFERENZE
DI
MOHR

$$\bar{C}_1, \bar{C}_2, \bar{C}_3 = 0$$

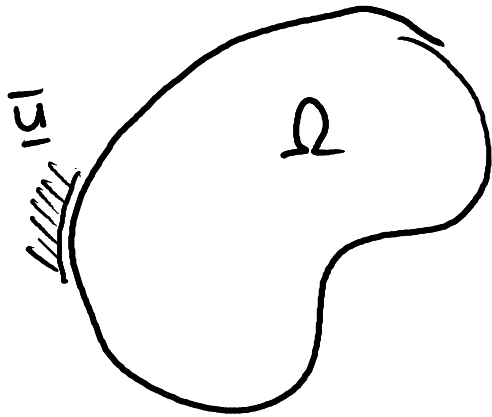
ARBELO
DI
MOHR

$$\begin{cases} \bar{C}_1 \geq 0 & \vee \\ \bar{C}_2 \leq 0 & \vee \\ \bar{C}_3 \geq 0 \end{cases}$$

EQUAZIONE DI CONGRUENZA

EQUAZIONE DI CONGRUENZA

Assegnare un campo di spostamento $\underline{u}(\underline{x})$ t.c. $\underline{u}(\underline{x}) \in C^1$ su Ω
 $\underline{u} = \bar{\underline{u}}$ su $\partial\Omega$



Le deformazioni dovute a questo campo sono

$$\epsilon_x = \frac{\partial u}{\partial x} \quad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

$$\epsilon_y = \frac{\partial v}{\partial y} \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$$

$$\epsilon_z = \frac{\partial w}{\partial z} \quad \gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$$

$$\underline{\underline{E}} = \text{sym}(\text{grad } \underline{u}) = \frac{1}{2}(\bar{\nabla} \underline{u} + \bar{\nabla}^T \underline{u})$$

EQUAZIONI DI CONGRUENZA ESTERNA
 (tra $\underline{u}$ e $\underline{\underline{E}}$)

Se si volesse assegnare un campo di deformazione $\epsilon_x, \epsilon_y, \epsilon_z, \gamma_{xy}, \gamma_{xz}, \gamma_{yz}$ come si potrebbe assicurare la congruenza, ossia la compatibilità tra queste deformazioni e la continuità del corpo?

$$\frac{\partial^2 \epsilon_x}{\partial y^2} + \frac{\partial^2 \epsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y}$$

$$\frac{\partial^2 \epsilon_x}{\partial z^2} + \frac{\partial^2 \epsilon_z}{\partial x^2} = \frac{\partial^2 \gamma_{xz}}{\partial x \partial z}$$

$$\frac{\partial^2 \epsilon_y}{\partial z^2} + \frac{\partial^2 \epsilon_z}{\partial y^2} = \frac{\partial^2 \gamma_{yz}}{\partial y \partial z}$$

$$2 \frac{\partial^2 \epsilon_x}{\partial y \partial z} = \frac{\partial}{\partial x} \left(\frac{\partial \gamma_{xy}}{\partial z} + \frac{\partial \gamma_{xz}}{\partial y} - \frac{\partial \gamma_{yz}}{\partial x} \right)$$

$$2 \frac{\partial^2 \epsilon_y}{\partial x \partial z} = \frac{\partial}{\partial y} \left(\frac{\partial \gamma_{xy}}{\partial z} + \frac{\partial \gamma_{yz}}{\partial x} - \frac{\partial \gamma_{xz}}{\partial y} \right)$$

$$2 \frac{\partial^2 \epsilon_z}{\partial x \partial y} = \frac{\partial}{\partial z} \left(\frac{\partial \gamma_{xz}}{\partial y} + \frac{\partial \gamma_{yz}}{\partial x} - \frac{\partial \gamma_{xy}}{\partial z} \right)$$

$$\text{rot}(\text{rot } \underline{\underline{E}}) = \bar{\nabla} \times (\bar{\nabla} \times \underline{\underline{E}}) = \underline{\underline{0}}$$

EQUAZIONI DI CONGRUENZA INTERNA
 (tra le componenti di $\underline{\underline{E}}$)